Law Office Solution Proposal

Toronto Metropolitan University
Ted Rogers School of Management

Group 8

ITM 900 - 011

Date: April 11, 2023

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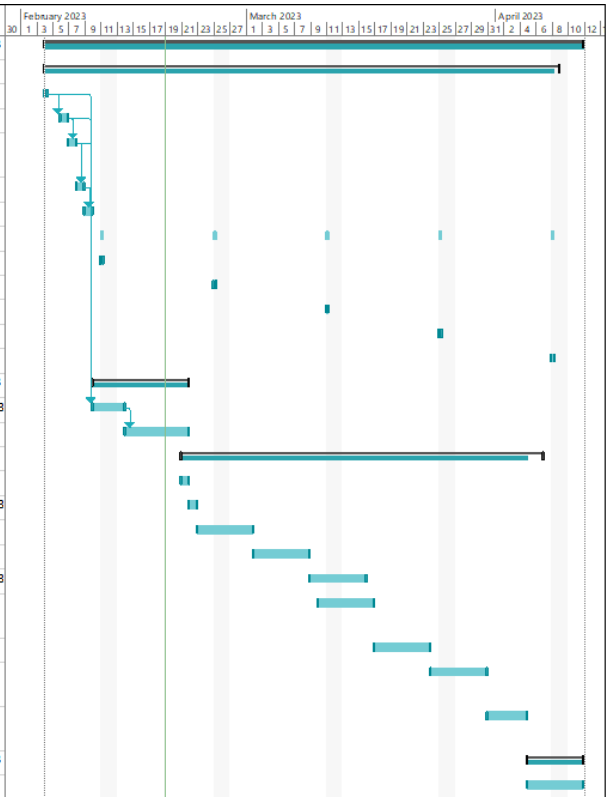
Project Management
Project Charter

Project Title: Law Office Solution Proposal	
Project Start Date: Feb 4th, 2023	**Projected Finish Date:** April 11th, 2023
Project Manager: Mohammad Shah	m1shah@torontomu.ca
Project Objectives: The Business owner wants to create a more efficient way to schedule meetings for potential clients. The current system relies on communication via email which requires too much wait time for responses resulting in a potential loss of clients, as they might search for alternative options. The objective of the project is to create a concrete proposal solution and prototype implementation of a system that can efficiently schedule meetings.	
Project Success Criteria: To create a solution proposal that can be implemented on the website https://www.hasinlaw.ca/ to increase efficiency for clients when booking an appointment and be an easily navigable system for clients and lawyers to use.	
Approach: - Meet with the client to identify their goals and what problems they face when reaching it and what opportunities could it bring to the firm - Develop project charter and project timeline detailing what a successful project result would be, how we will tackle the problem, and how long it will take. - Meet with the client to gather a more in-depth understanding of their current process. - Create a solution that will target the problem that the client is facing. - Schedule weekly zoom meetings with team members to go over progress and ensure we are on track. - Schedule bi-weekly meetings with our client to provide updates and receive feedback on the solution being developed.	
Roles and Responsibilities	

Name	***Role***	***Position***	***Contact***
Mohammad Shah	Project Manager	Student	
Andy Eng	Team Member	Student	
Tommy Fan	Team Member	Student	
Hasin Khan	Team Member	Student	
Ravneet Basi	Team Member	Student	
Hasin Chaudhary	**Business Owner**	**Client**	

Project Timeline (Gantt Chart)

ID	Task Mode	Task Name	Duration	Start	Finish
1		**Law Office Solution Proposal Project**	**49 days**	**Sat 2/4/23**	**Tue 4/11/23**
2		**Initiation and Discussion**	**48 days**	**Sat 2/4/23**	**Sat 4/8/23**
3		Kickoff Meeting with Clients	1 hr	Sat 2/4/23	Sat 2/4/23
4		Identify Problems, Issues, and Opportunities	1 day	Mon 2/6/23	Mon 2/6/23
5		Discuss Project Requirements, Purpose, and Plan	1 day	Tue 2/7/23	Tue 2/7/23
6		Gather Requirements	1 day	Wed 2/8/23	Wed 2/8/23
7		Decide Scope	1 day	Thu 2/9/23	Thu 2/9/23
8		Bi-Weekly Update Meetings with Client	0.13 days	Sat 4/8/23	Sat 4/8/23
9		Bi-Weekly Update Meetings with Client 1	1 hr	Sat 2/11/23	Sat 2/11/23
10		Bi-Weekly Update Meetings with Client 2	1 hr	Sat 2/25/23	Sat 2/25/23
11		Bi-Weekly Update Meetings with Client 3	1 hr	Sat 3/11/23	Sat 3/11/23
12		Bi-Weekly Update Meetings with Client 4	1 hr	Sat 3/25/23	Sat 3/25/23
13		Bi-Weekly Update Meetings with Client 5	1 hr	Sat 4/8/23	Sat 4/8/23
14		**Planning**	**9 days**	**Fri 2/10/23**	**Tue 2/21/23**
15		Create Draft of Project Plan of Action	2 days	Fri 2/10/23	Mon 2/13/23
16		Propose Project Plan of Action to Client	7 days	Tue 2/14/23	Tue 2/21/23
17		**Executing**	**33 days**	**Tue 2/21/23**	**Thu 4/6/23**
18		Create Project Charter	1 day	Tue 2/21/23	Tue 2/21/23
19		Create Project Schedule	1 day	Wed 2/22/23	Wed 2/22/23
20		Requirements Specification & Modelling	5 days	Thu 2/23/23	Wed 3/1/23
21		Business Process Modelling	5 days	Thu 3/2/23	Wed 3/8/23
22		Data Modelling & Design of the New IS Solution	5 days	Thu 3/9/23	Wed 3/15/23
23		Structual & Behavioral Design of the New IS Solution	5 days	Fri 3/10/23	Thu 3/16/23
24		Proof-of-Concept Implementation Prototype	5 days	Fri 3/17/23	Thu 3/23/23
25		Implementation and Deployement & Risk Management Plan for the New IS Solution	5 days	Fri 3/24/23	Thu 3/30/23
26		Prepare Final Presentation to Client and Professor	3 days	Fri 3/31/23	Tue 4/4/23
27		**Close**	**5 days**	**Wed 4/5/23**	**Tue 4/11/23**
28		Review Final Project	5 days	Wed 4/5/23	Tue 4/11/23



Identification of the Problem(s)/Opportunities

Problem:

At the Hasin Chaudhary law office, our client faces time management issues and delays when it comes to scheduling meetings with potential clients of their own. The current scheduling process relies solely on communication via email which requires one to two business days for responses from both parties.The current method leads to more issues such as double booking which can cause inconveniences to all parties involved. This has led to a loss of clients due to delays from the result of long wait times between responses and excessive emailing to determine an optimal time to schedule a meeting. The law office also deals with issues arising from clients not providing all the necessary information ahead of time via email before the meetings resulting in the lawyer taking up more time during the meeting just to grasp the situation rather than being able to immediately get into what support and advice they can offer. This has led to dissatisfaction from both the business owner and their potential clients leading to a lower rate of return for these potential clients. The current process is not an optimal method for a law office and the law office wants a more efficient means for both scheduling and managing appointments which would allow for their potential clients to book appointments with ease and with minimal back and forth between parties.

Opportunity:

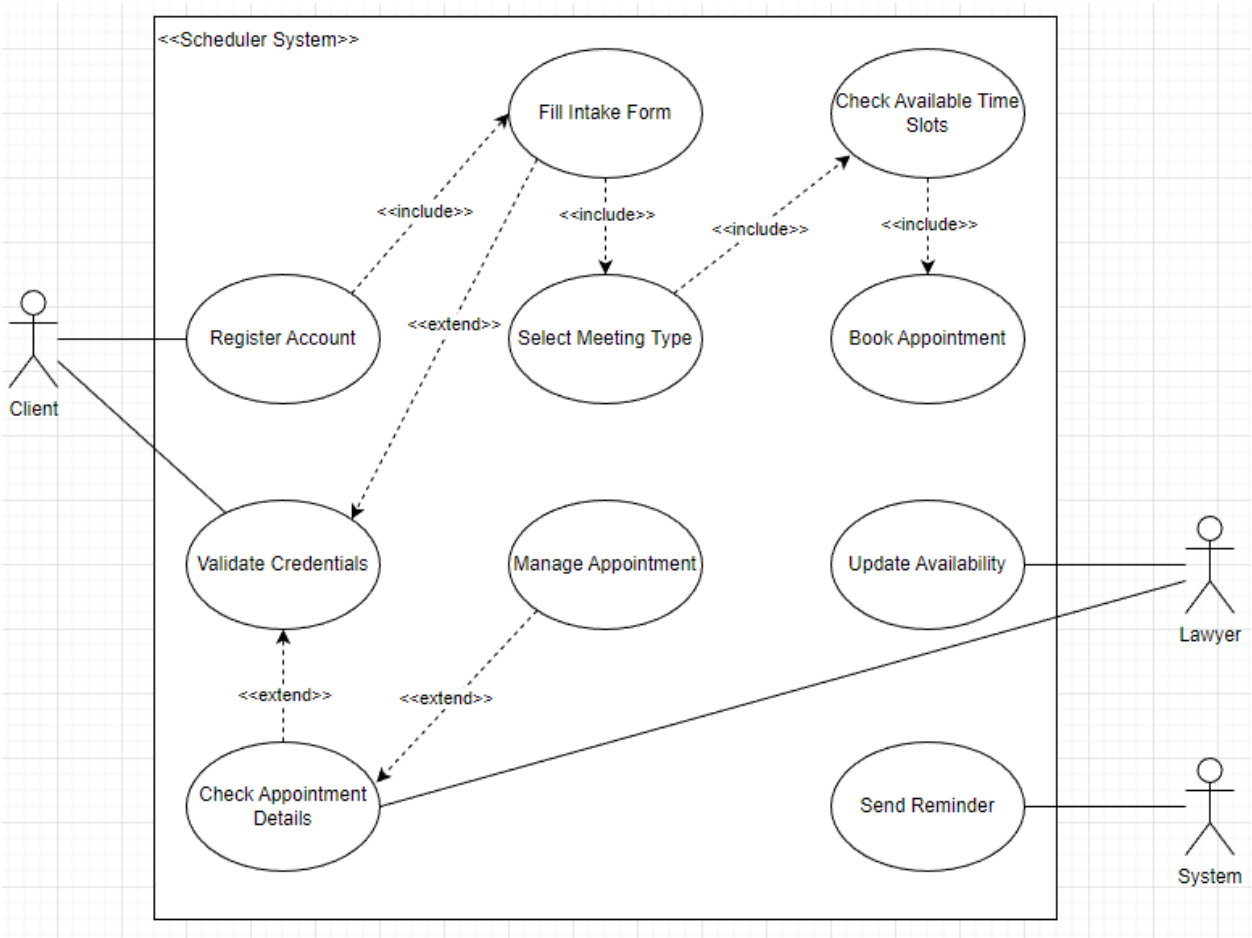
A solution to the given problem listed above will bring many opportunities to the business. An opportunity to help fix the problem could be to improve the already existing website by introducing a scheduling system that would streamline the action of booking appointments to reduce wait time and remove the risk of double booking. By utilizing this system it will remove pressure from the law office to have to manually set up appointments and organize between themself and their potential clients and having to ensure that they have all the necessary details before the meeting. To use the system to book an appointment, the client would have to register for an account on the website. Once registered and logging into their account the client will have to fill out an intake form with their personal information and details about the case to ensure the lawyer will have an understanding of the situation before the meeting. After completing the intake form the client will select their preferred meeting length, form of communication, date and time from the available options before confirming their appointment. An email reminder will be sent out two days from the appointment and on the day of to help ensure the client and lawyer attend the meeting on time. This process will give all parties ease of use and remove any uncertainties when deciding on a meeting date and the uncertainty of if one of the parties has forgotten or is unaware of the scheduled meeting.

Diagnostic (Causes-Problem-Consequences)

Causes	Problem	Consequences
Organizational issue caused by receiving all appointment requests by email.	Current scheduling process creates risk for overlapping appointments and lost emails.	Clients will be dissatisfied with rescheduled appointments due to double booking or not receiving a response at all.
Law office does not respond to appointment emails fast enough.	Clients have to wait for a response from the lawyer before they can proceed.	Clients may get irritated or have an urgent problem that needs to be addressed and move to other firms due to slow communication.
Clients are not providing the necessary information regarding the case beforehand.	Lawyer takes up a significant portion of the meeting trying to understand the case rather than being able to provide useful support and advice from the start.	Clients will be dissatisfied that their meeting is being spent unproductively and may cause them to think that they are not receiving adequate services.
Current website not optimized.	Current website has no login system to confirm if clients are real.	The law office will be swamped with emails from possible malicious actors resulting in emails from potential clients to be overlooked.

Requirement Specification & Modeling

Use Case Model



Use Case Descriptions:

Register Account

Use Case	Register Account
Actors	Client
Short Description	Client creates an account by filling in required information
Trigger	Client wants to register for an account
Preconditions	Client is not already registered
Normal Flow (Client creates a new account)	1. Client goes to the login/register page 2. System prompts client if they have an account 3. Client enters their email 4. Client enters a password 5. Client confirms the password 6. Account is registered
Alternative Flow	N/A
Exceptions	1. Client has an existing account 2. Client does not fill in all information
Post Conditions	Account is successfully created. Client is redirected to fill in an intake form
Includes	Fill Intake Form

Fill Intake Form

Use Case	Fill Intake Form
Actors	Client
Short Description	Client fills out intake form with information for the meeting
Trigger	Client wants to book an appointment
Preconditions	Client is signed into an account
Normal Flow (Client fills out the intake form)	1. Client fills out intake form 2. Client enters personal information 3. Client describes the legal matter and case 4. Client attaches necessary documents 5. System verifies the intake form is complete
Alternative Flow	N/A
Exceptions	Client is unable to fill out intake form
Post Conditions	Intake form is completed. Client is redirected to select their meeting preferences
Includes	Select Meeting Type

Select Meeting Type

Use Case	Select Meeting Type
Actors	Client
Short Description	The Client selects their preferred meeting length and form of communication
Trigger	Client has completed the intake form
Preconditions	Intake form is complete
Normal Flow (Client chooses the length of the meeting and form of communication)	1. Client selects meeting length 2. Client selects form of communication
Alternative Flow	N/A
Exceptions	Client does not select any options
Post Conditions	Appointment meeting length and form of communication is confirmed. Client is redirected to select an available time slot.
Includes	Check Available Time Slots

Check Available Time Slots

Use Case	Check Available Time Slots
Actors	Client
Short Description	Client views available dates and times from which they can book an appointment with the lawyer
Trigger	Client has selected a meeting length and form of communication
Preconditions	Meeting length and form of communication are selected
Normal Flow (Client selects an available time slot)	1. Client views available dates 2. Client selects an available meeting date and time
Exceptions	There are no desired time slots
Post Conditions	Client has selected an available time slot. Client is redirected to confirm appointment details
Includes	Book Appointment

Book Appointment

Use Case	Book Appointment
Actors	Client
Short Description	Client reviews all appointment details before confirming the appointment
Trigger	Client has chosen an available time slot for their appointment
Preconditions	Available time slot was chosen
Normal Flow (Confirm appointment)	1. Client reviews the intake form 2. Client reviews the selected meeting date, time and form of communication 3. Client confirms appointment details 4. System updates time slots
Alternative Flow (Mistakes are found in appointment details)	1. Client reviews the intake form 2. Client reviews the selected meeting date, time and form of communication 3. Client returns to previous steps to correct mistakes
Exceptions	Client changes their mind about booking an appointment and exits
Post Conditions	Client receives an email confirmation for their booking
Includes	N/A

Update Availability

Use Case	Update Availability
Actors	Lawyer
Short Description	Lawyer updates time slot availability
Trigger	Lawyer wants to update their availability
Preconditions	Lawyer is logged into the admin account
Normal Flow (Lawyer updating time slot availability)	1. Lawyer selects a time slot 2. Lawyer updates the availability of the time slot
Exceptions	Lawyer does not change any time slots
Post Conditions	Availability of time slots are updated
Includes	N/A

Validate Credentials

Use Case	Validate Credentials
Actors	Client
Short Description	Customer uses their information to sign-in and system confirms account is registered
Trigger	Client wants to sign-in to their account
Preconditions	Account must be registered
Normal Flow (Client signs in)	1. System prompts client for email and password 2. System checks to verify that the account is registered 3. Client is signed into their account
Alternative Flow (Client makes too many sign in attempts)	1. System prompts client for email and password 2. System checks to verify that the account is registered 3. Client fails to sign in 3 times 4. System prompts Client to reset their password
Exceptions	Sign in information is incorrect
Post Conditions	Client is redirected to the account home page
Includes	Check Appointment Details

Check Appointment Details

Use Case	Check Appointment Details
Actors	Client, Lawyer
Short Description	Client and Lawyer are both able to view details of their appointments including date, time, and a brief description of the reason for the appointment.
Trigger	Client or Lawyer login into their account
Preconditions	Client is logged in. Lawyer is using the admin account
Normal Flow (Client views their appointment details)	1. Client/Lawyer selects the appointment they wish to view 2. System displays appointment details
Alternative Flow (Client desires to change appointment)	1. Client selects the appointment they wish to view 2. System displays appointment details 3. Client chooses to make changes to their booked appointment
Exceptions	N/A
Post Conditions	Client is redirected to the manage appointment menu if they want to make changes to their appointment
Includes	N/A

Manage Appointment

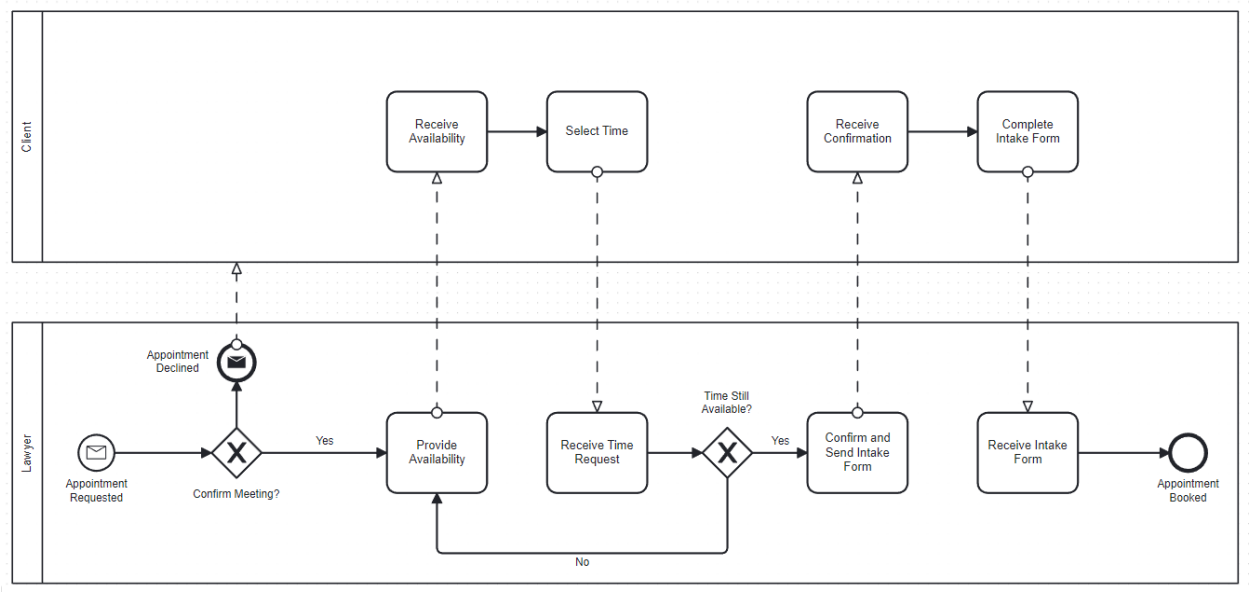
Use Case	Manage Appointment
Actors	Client
Short Description	Client will have the ability to either cancel or reschedule their appointment or to attach additional documents
Trigger	Client clicks on manage appointment
Preconditions	Client has an appointment booked
Normal Flow (Client wants to reschedule their appointment)	1. Client chooses to reschedule their booked appointment 2. System redirects the Client to choose a new time slot 3. Client chooses a new time slot 4. System cancels the old appointment 5. System updates time slots 6. System emails Client and Lawyer about the rescheduled appointment
Alternative Flow A (Client wants to cancel their appointment)	1. Client chooses to cancel their booked appointment 2. System cancels their appointment 3. System updates time slots 4. System emails Client and Lawyer about the appointment cancellation
Alternative Flow B (Client wants to attach additional documents)	1. Client chooses to attach additional documents to their appointment 2. System emails Client and Lawyer that the documents have been changed
Exceptions	No changes are made
Post Conditions	● Client is redirected to choose a new time slot if they want to reschedule their appointment ● System updates time slot availability and sends email if an appointment is canceled ● System sends emails if documents are updated
Includes	N/A

Send Reminder

Use Case	Send Reminder
Actors	System
Short Description	System sends an email to remind the Client and Lawyer of an upcoming appointment
Trigger	An appointment is scheduled in 2 days An appointment is scheduled today
Preconditions	Appointment is not canceled
Normal Flow	1. System checks if there are any appointments scheduled in 2 days 2. System checks if there are any appointments scheduled for today 3. System sends email reminders to the Clients and the Lawyer for applicable appointments
Alternative Flow	N/A
Exceptions	There are no upcoming appointments
Post Conditions	Email reminders are sent
Includes	N/A

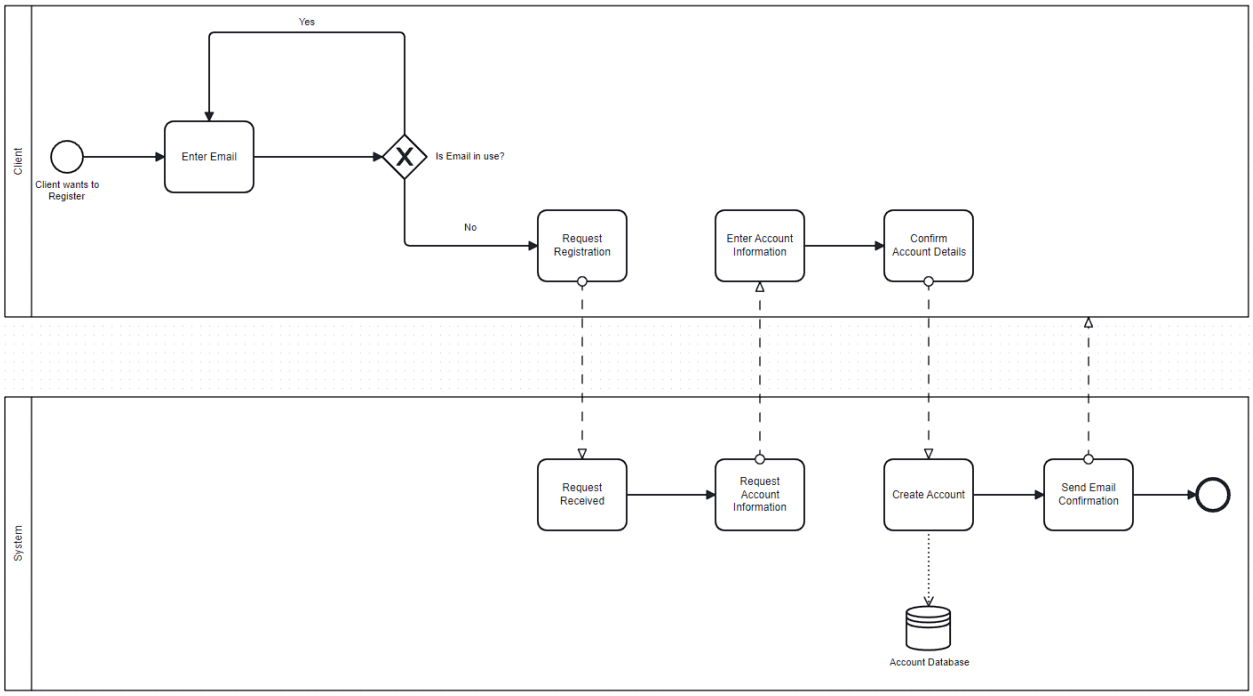
Business Process Modeling

<u>As-Is BPMN Model</u>

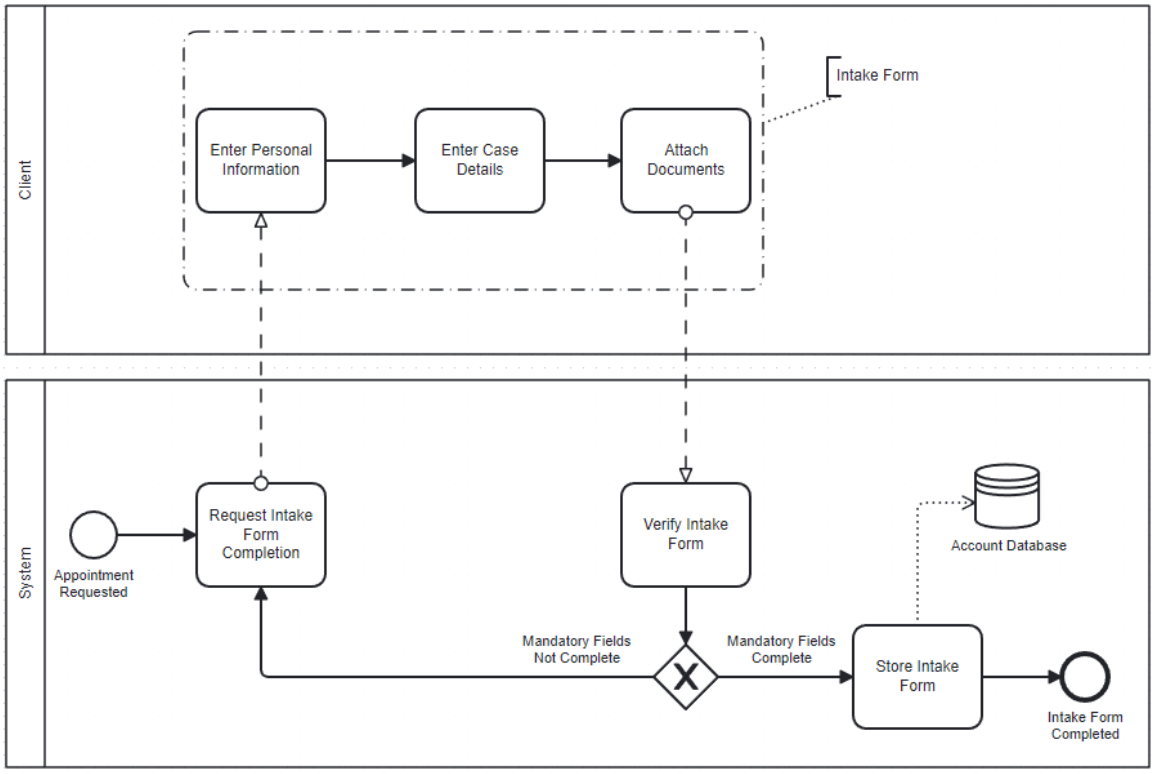


To-Be BPMN Model

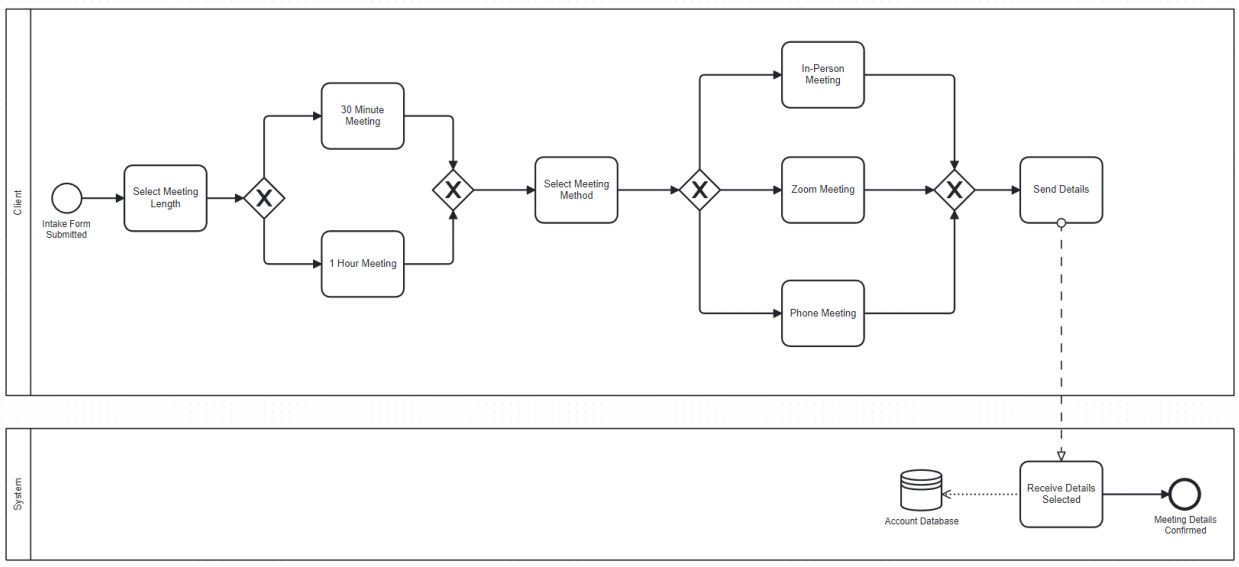
Register Account



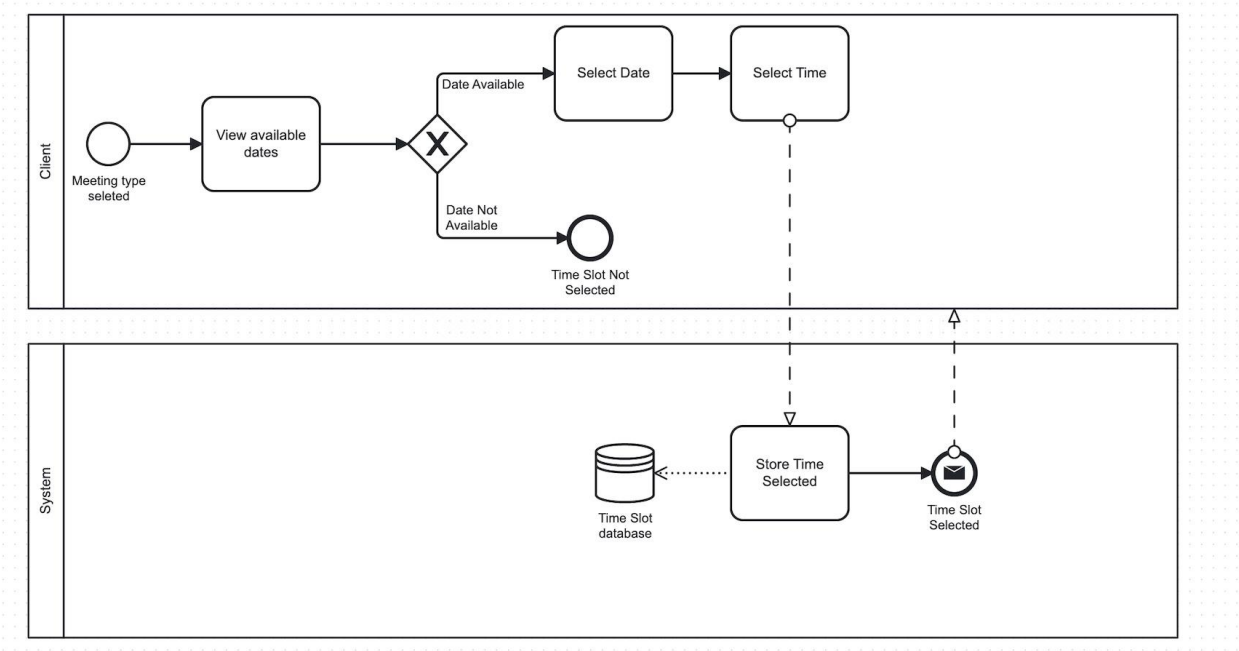
Fill Intake Form



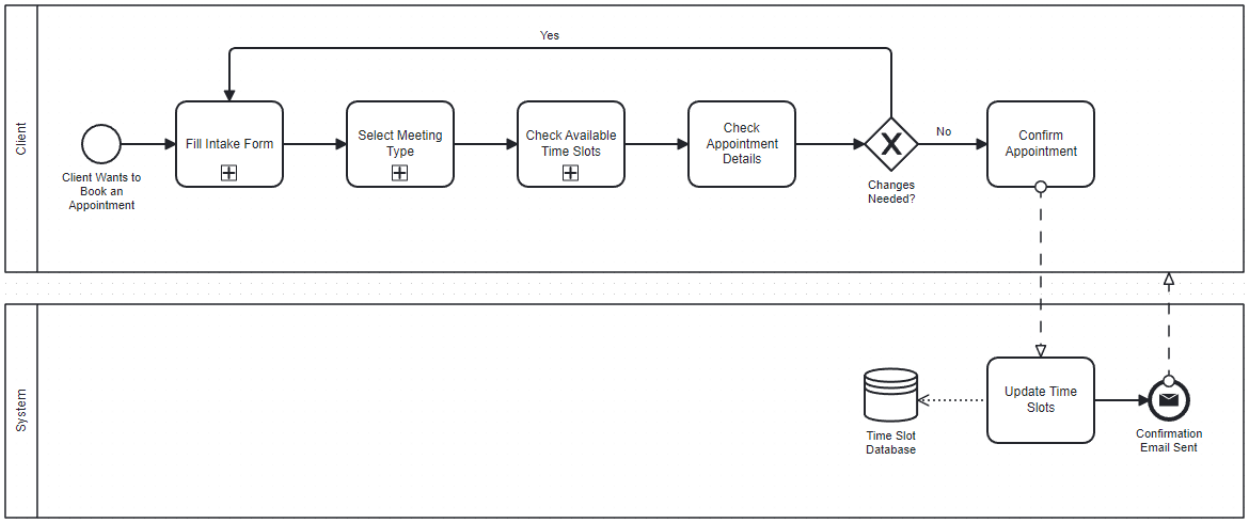
Select Meeting Type



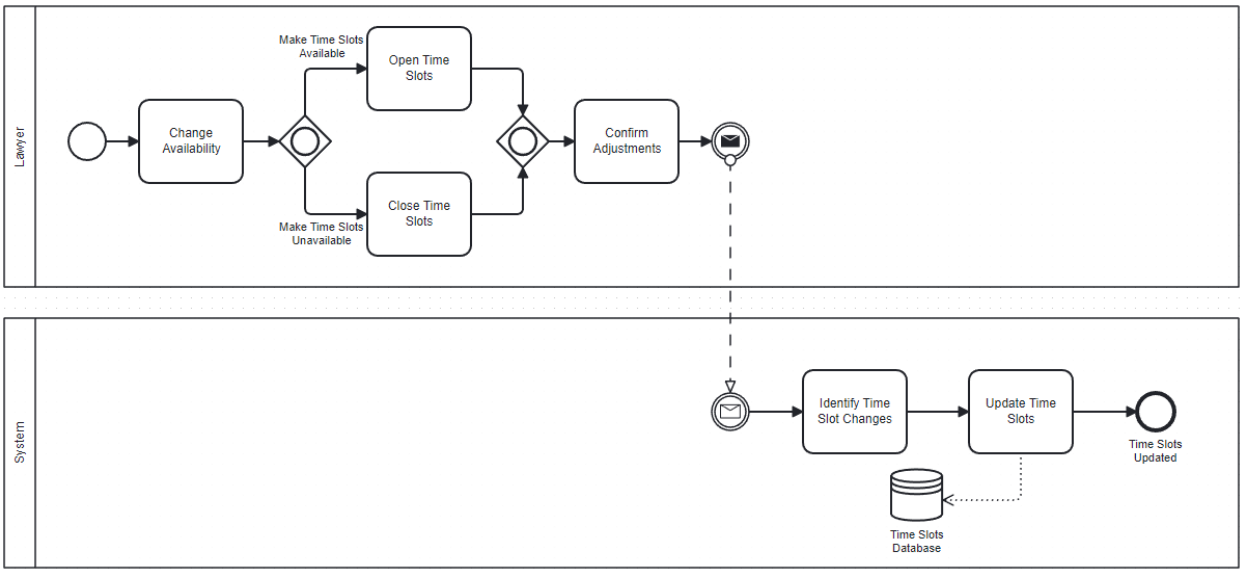
Check Available Time Slots



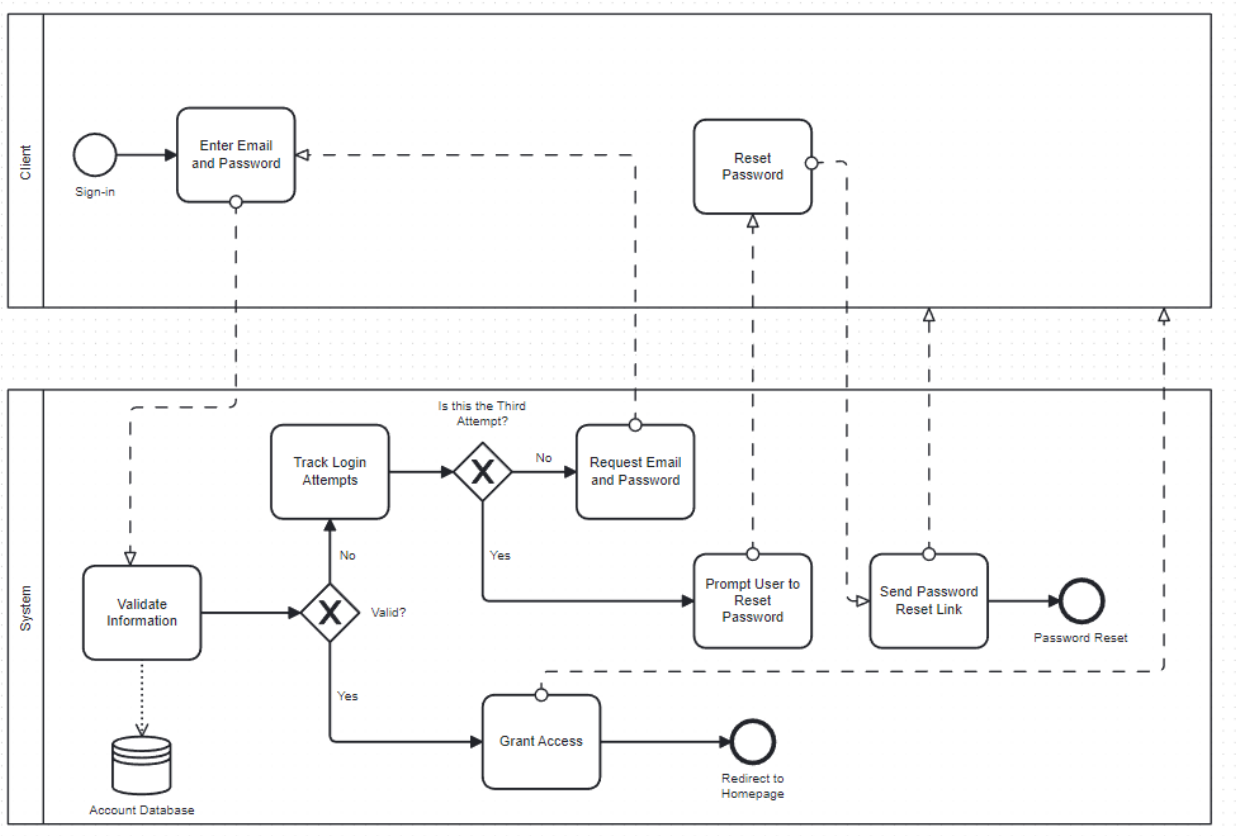
Book Appointment



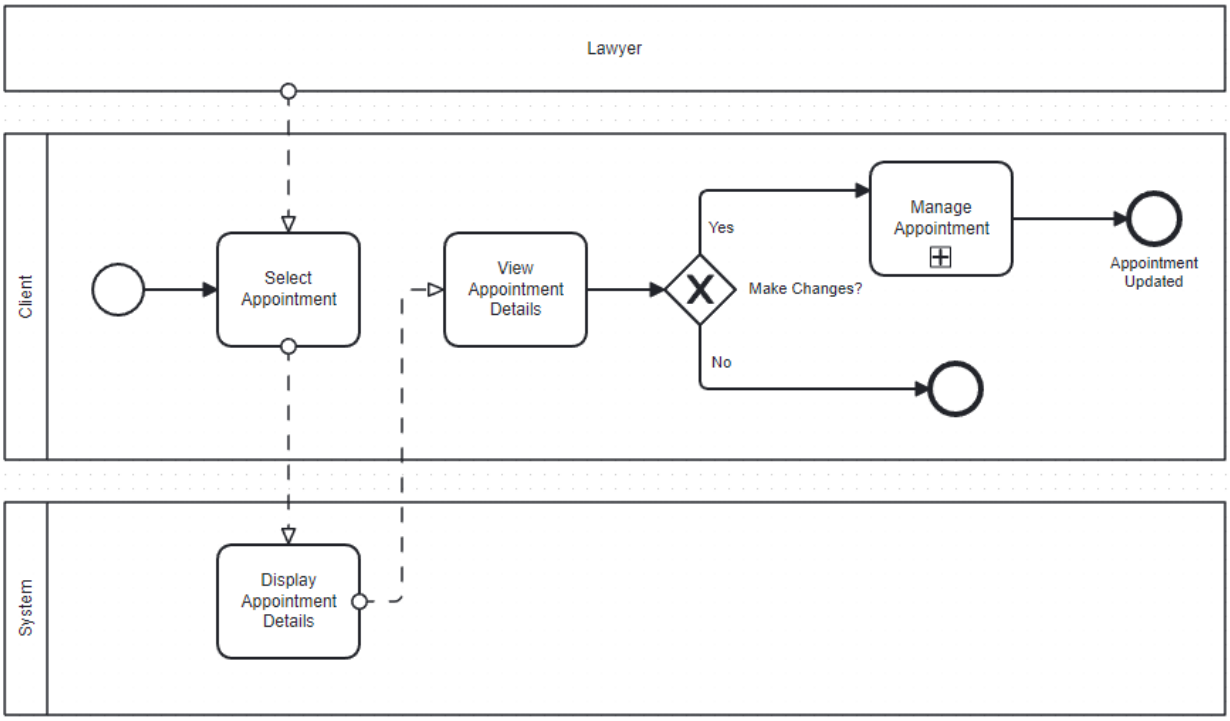
Update Availability



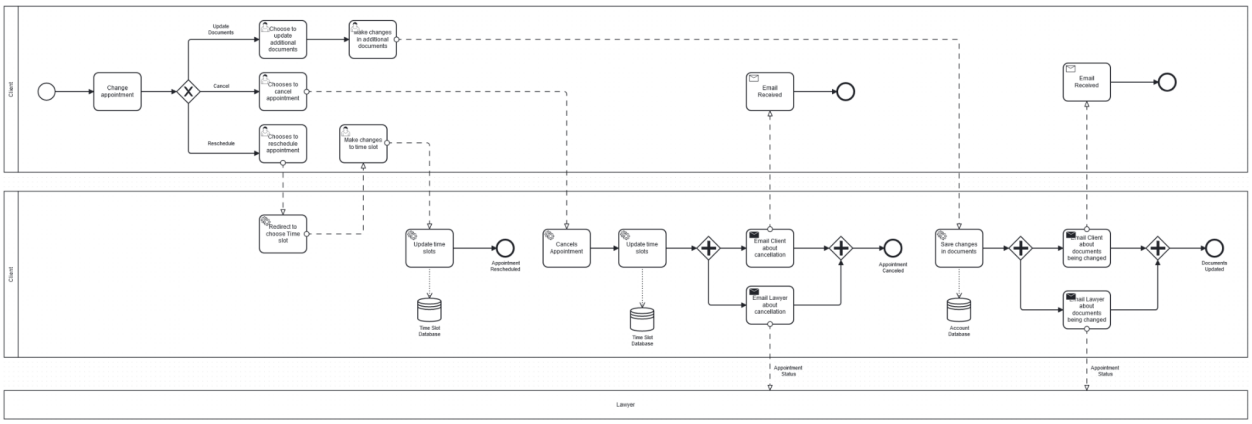
Validate Credentials



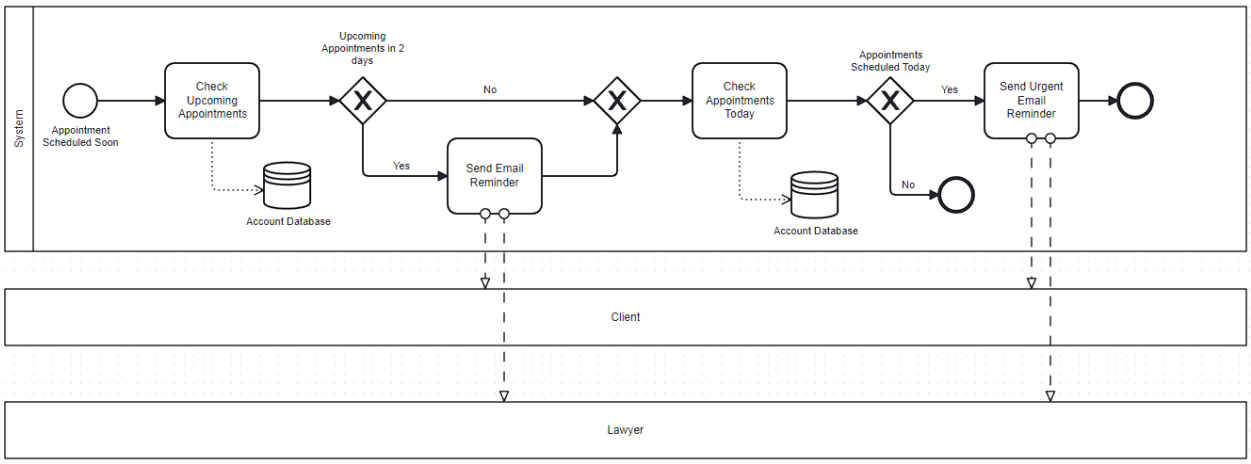
Check Appointment Details



Manage Appointment

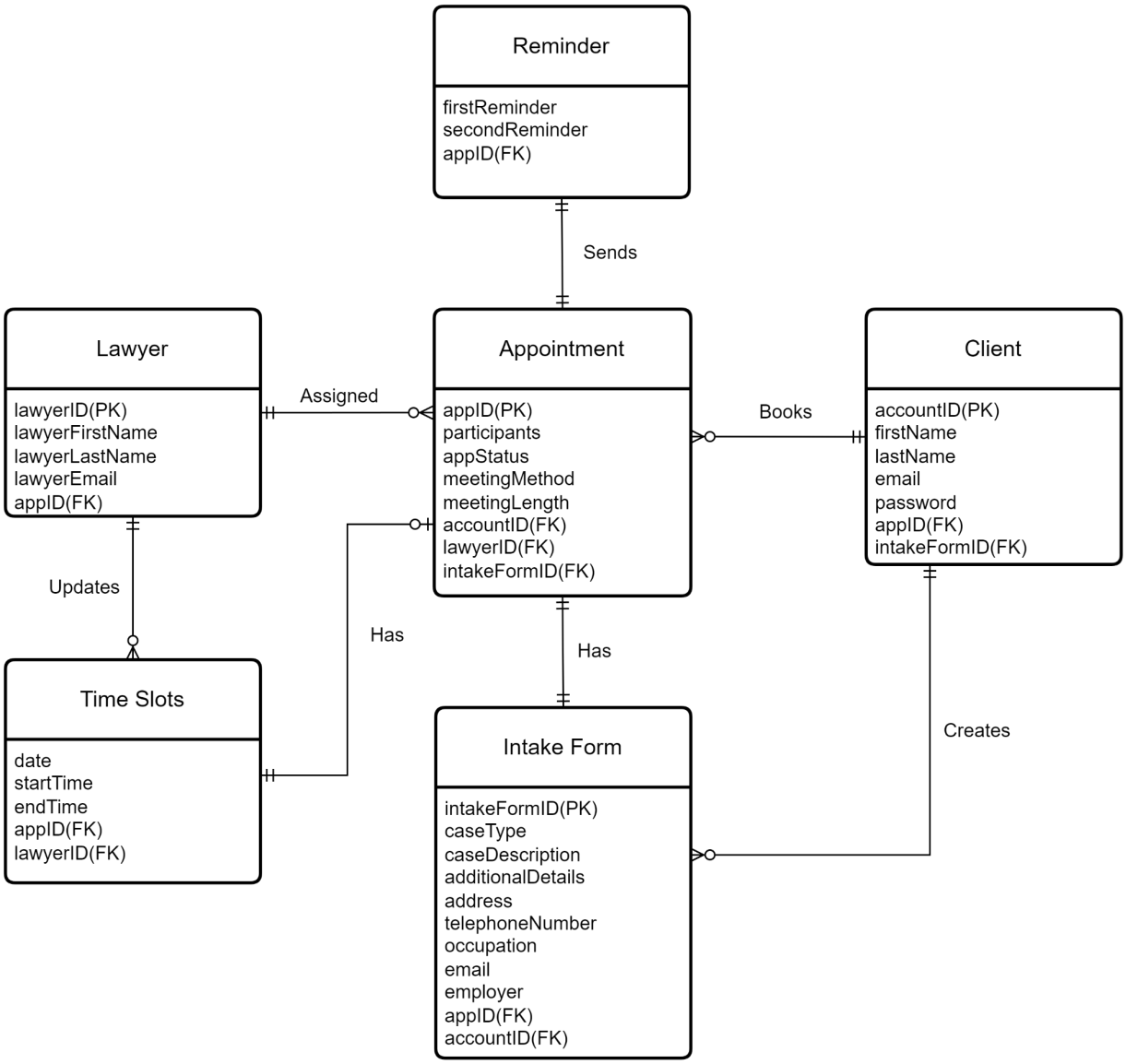


Send Reminder

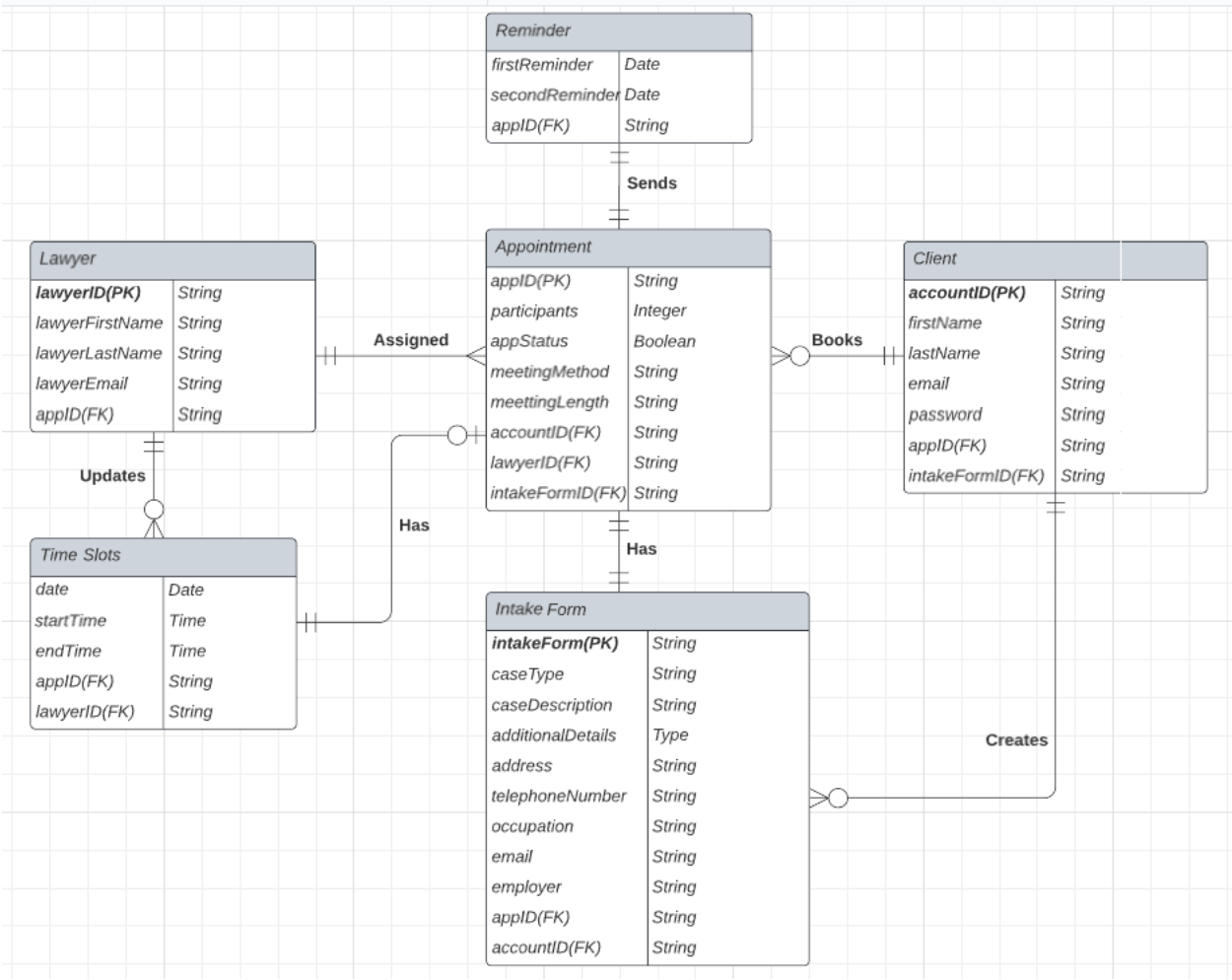


Data Modeling & Design of the new IS solution

Entity Relationship Diagram

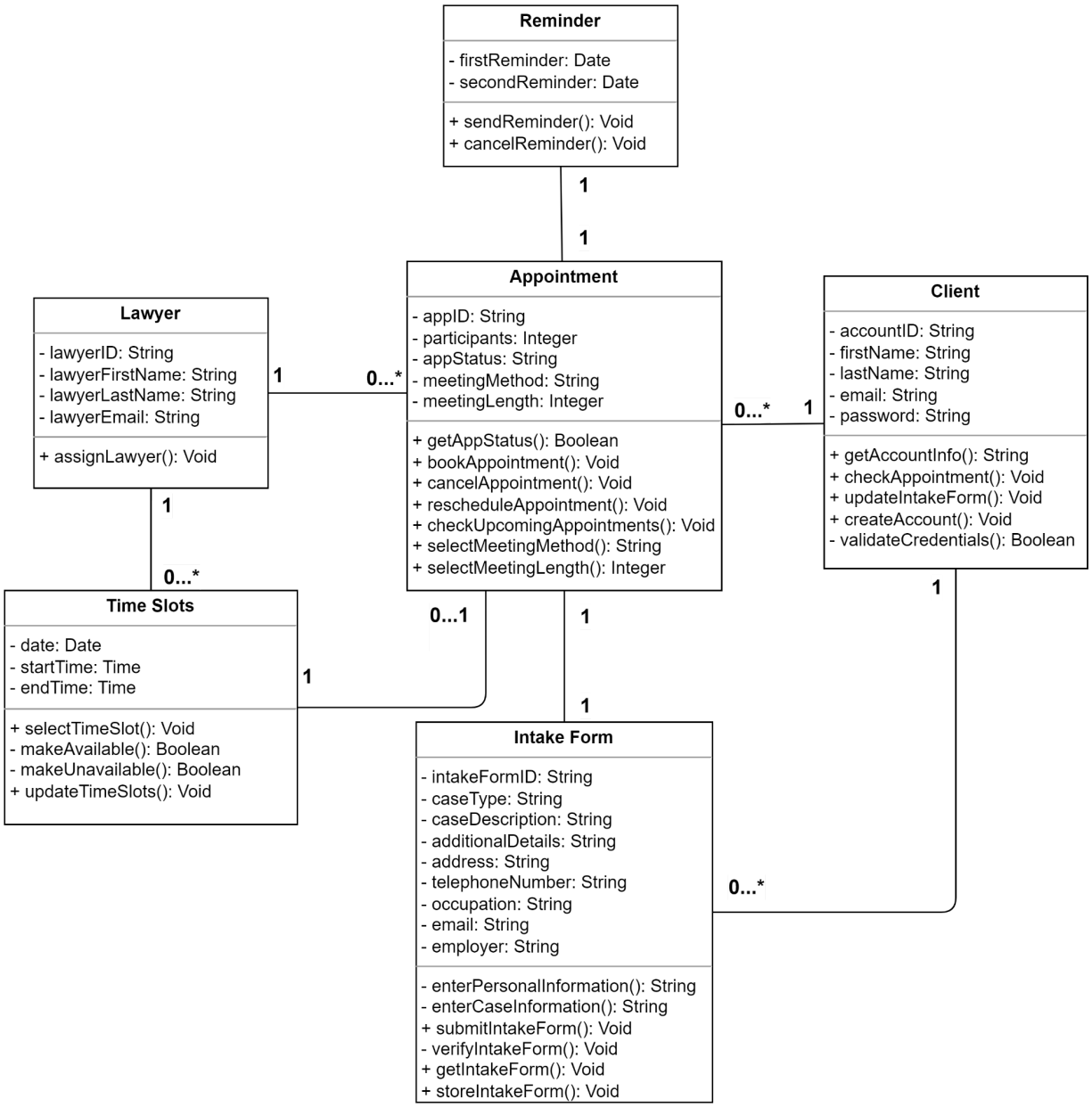


Database Schema



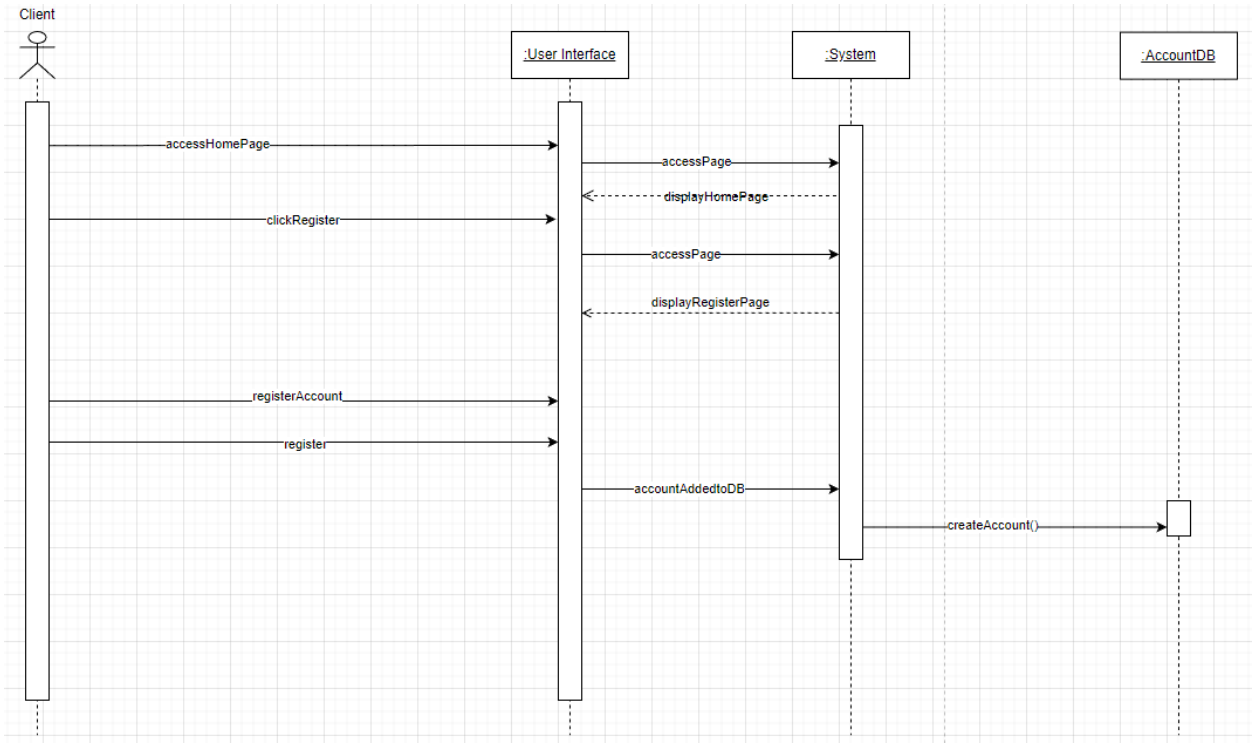
Structural & Behavioral Design of the New IS Solution

Class Diagram

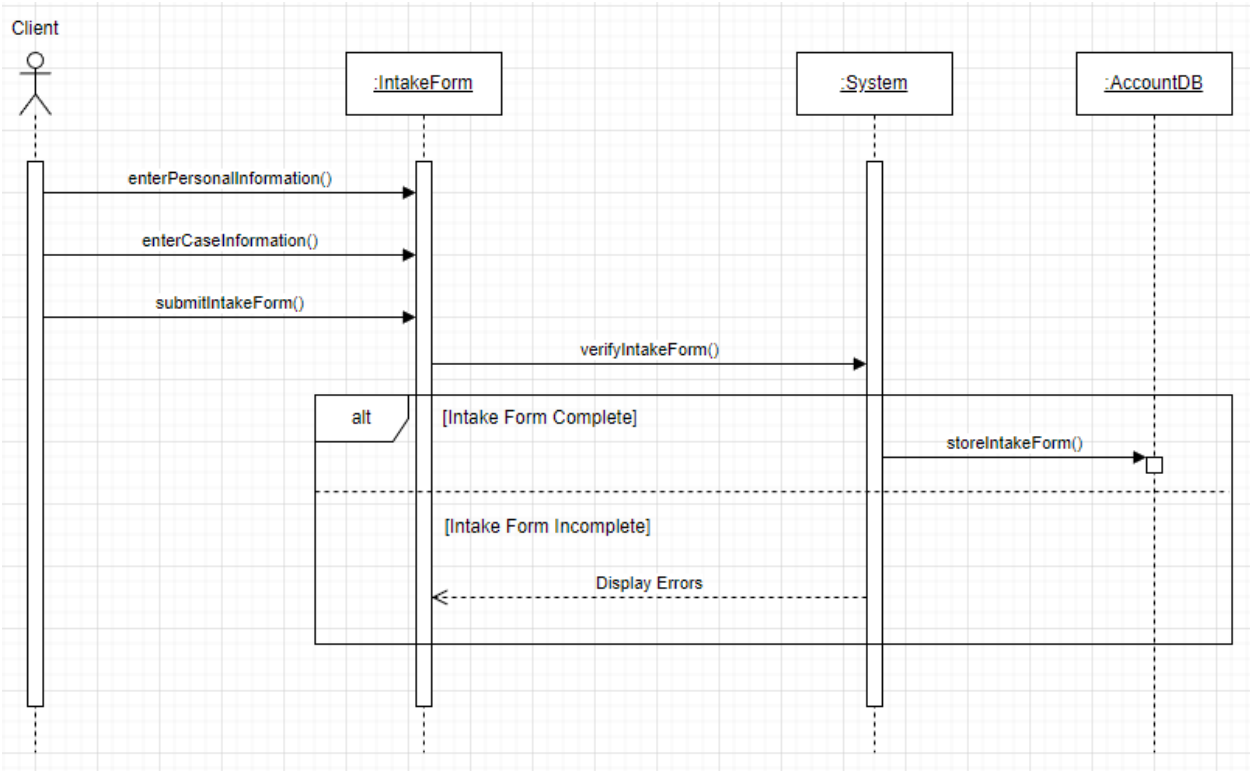


Sequence Diagrams:

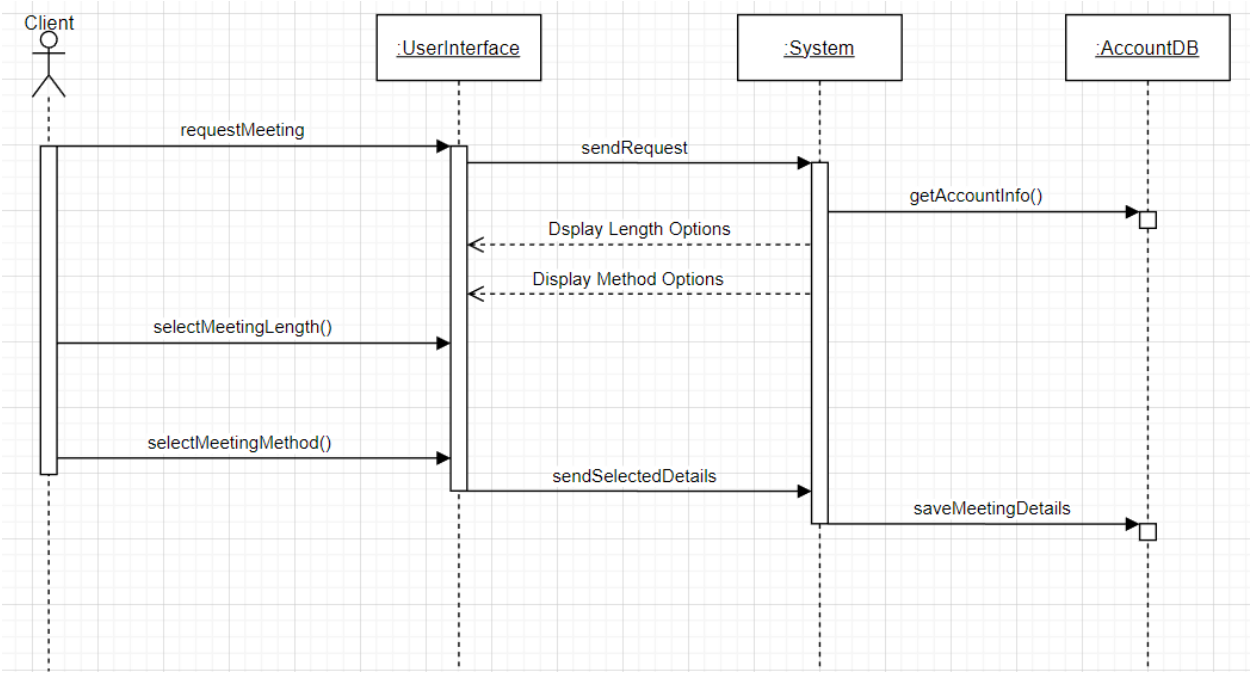
Register Account



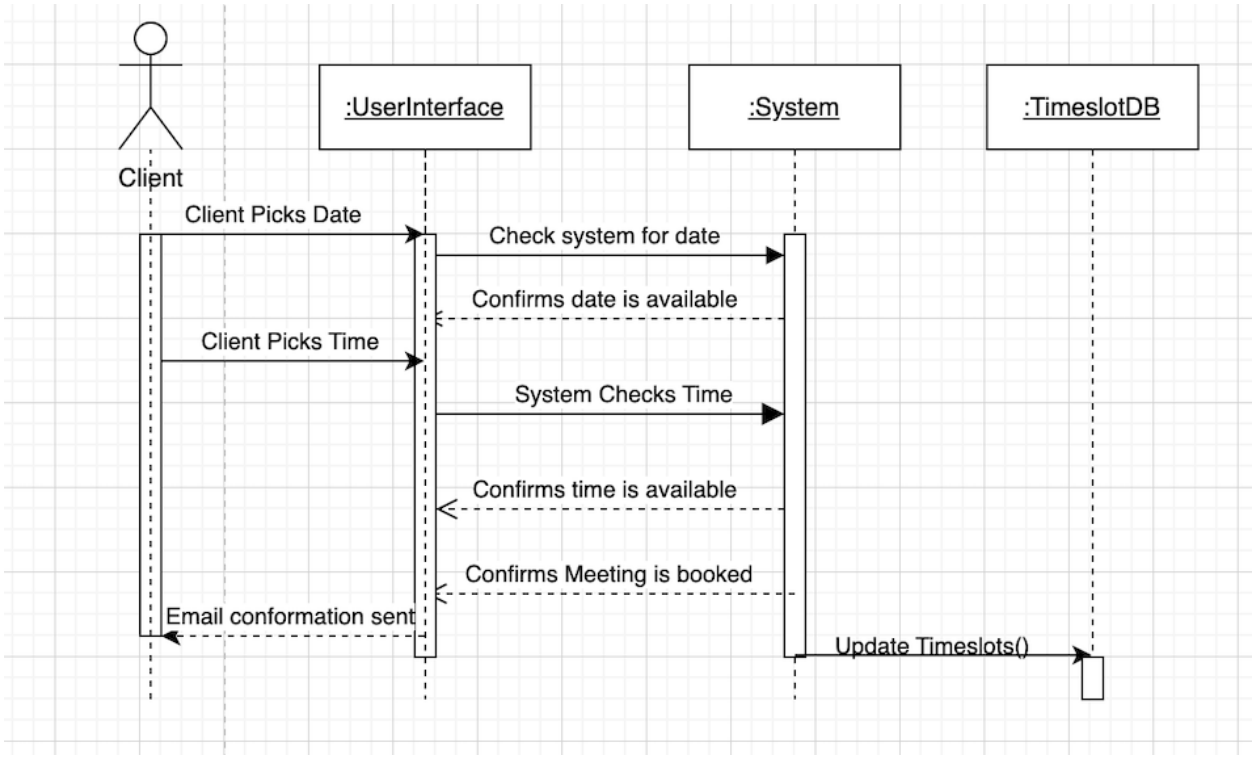
Fill Intake Form



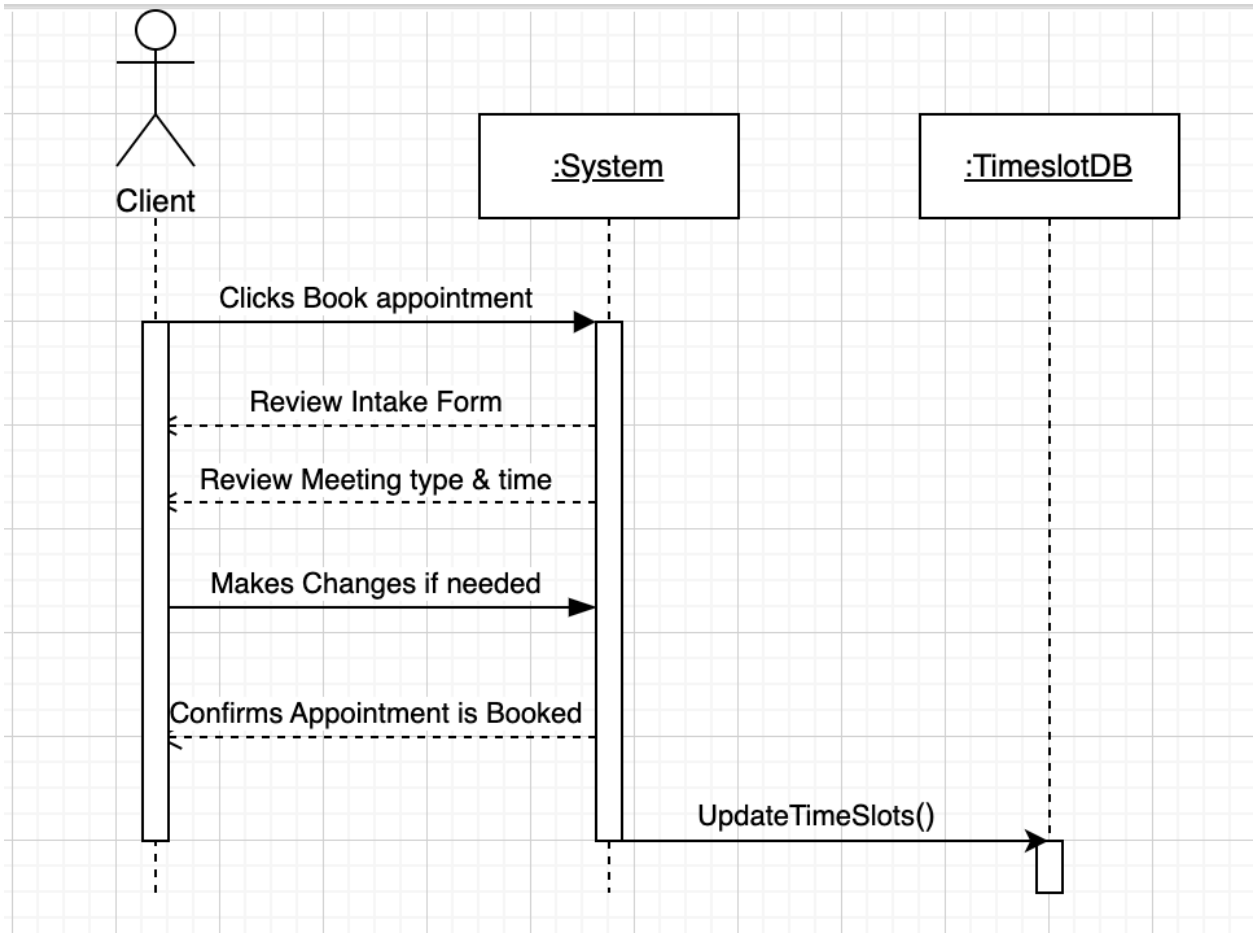
Select Meeting Type



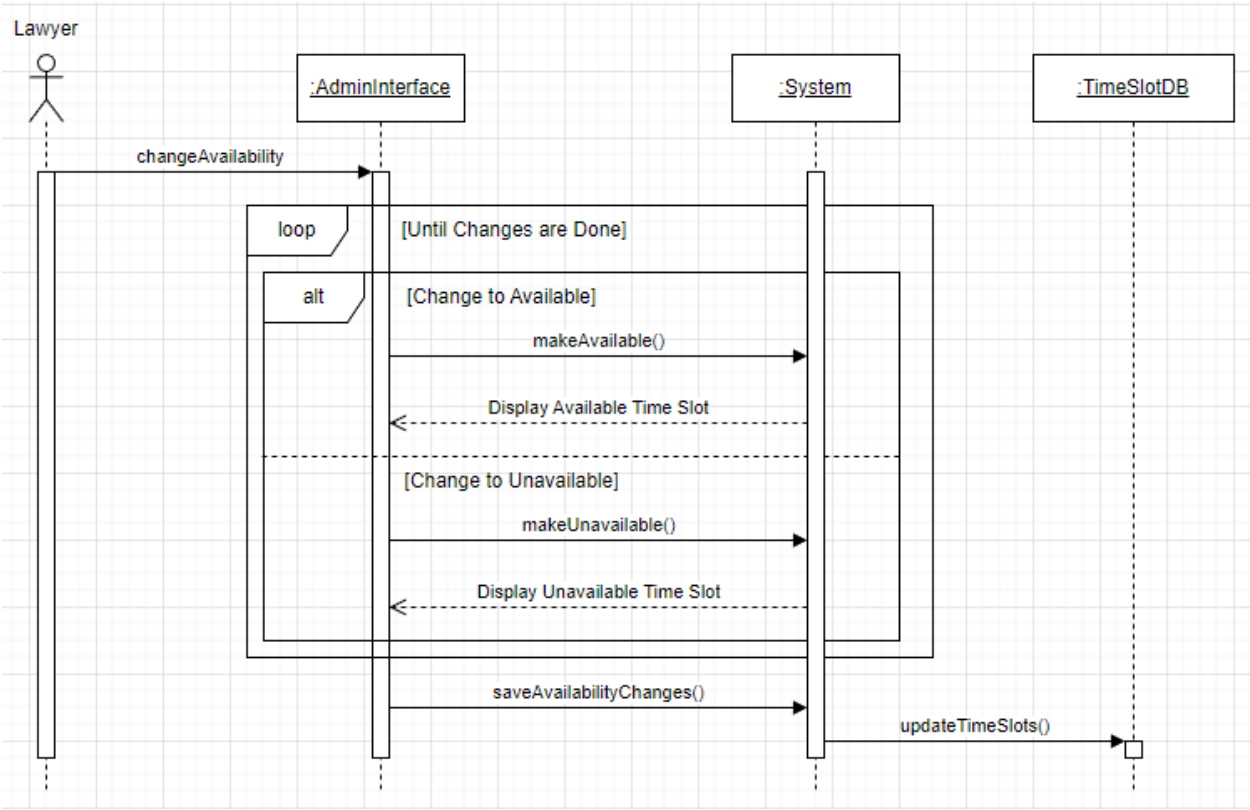
Check Available Time Slots



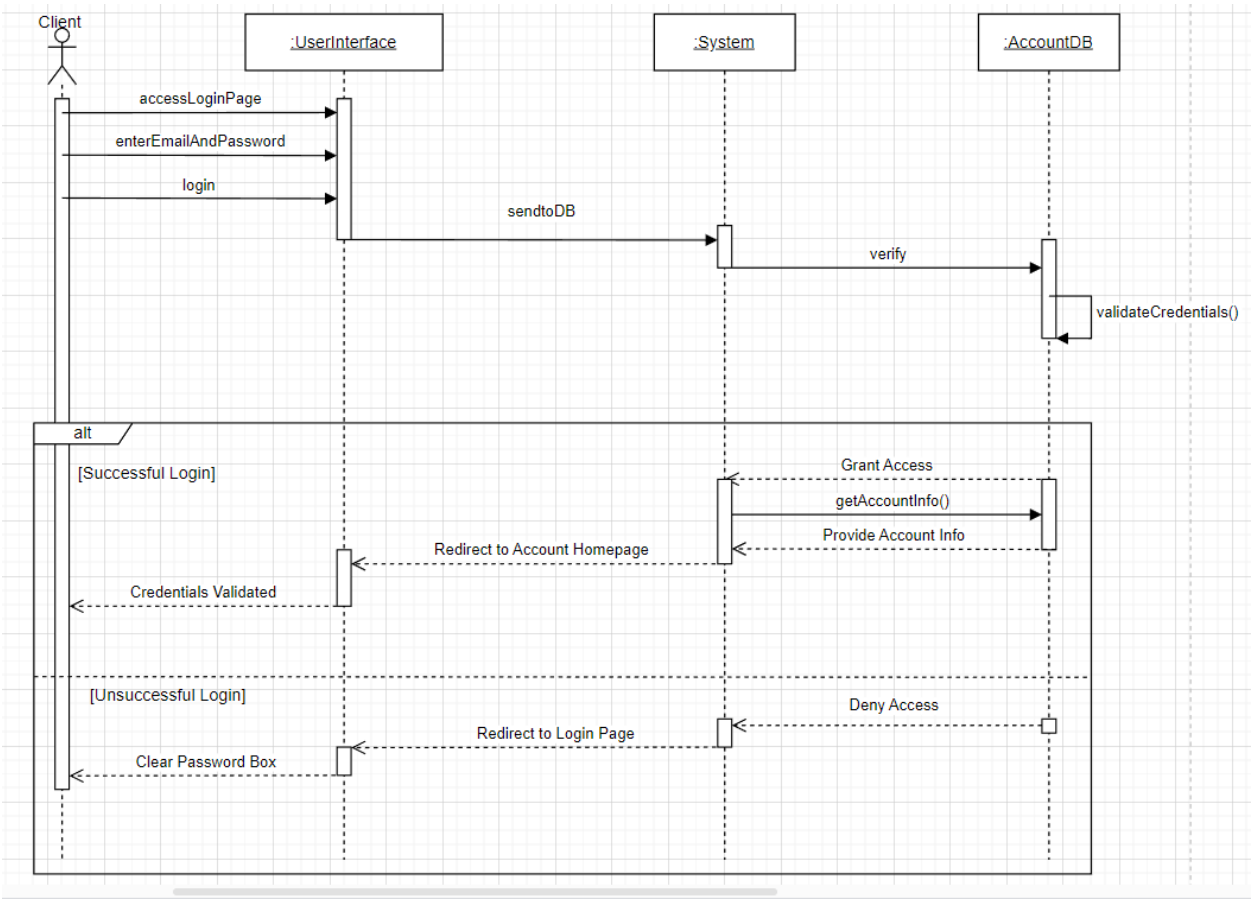
Book Appointment



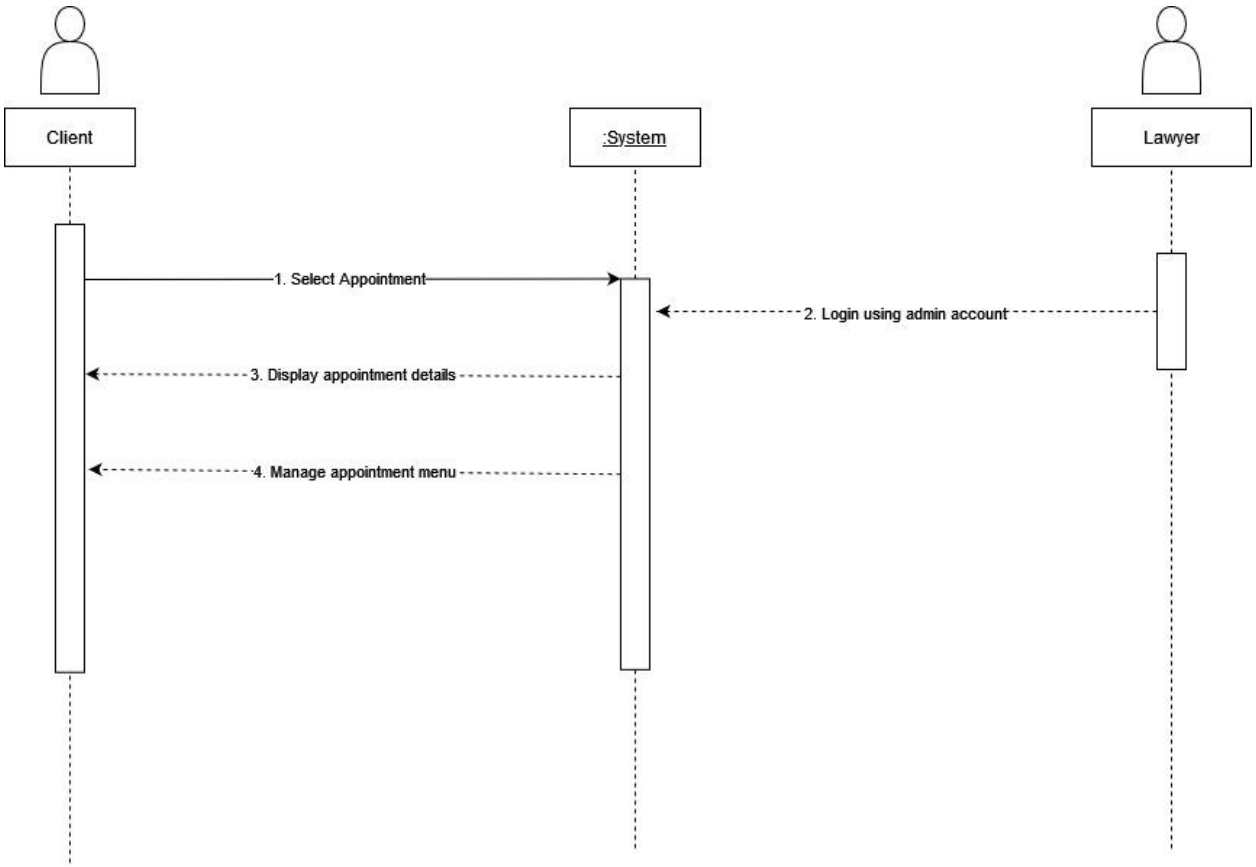
Update Availability



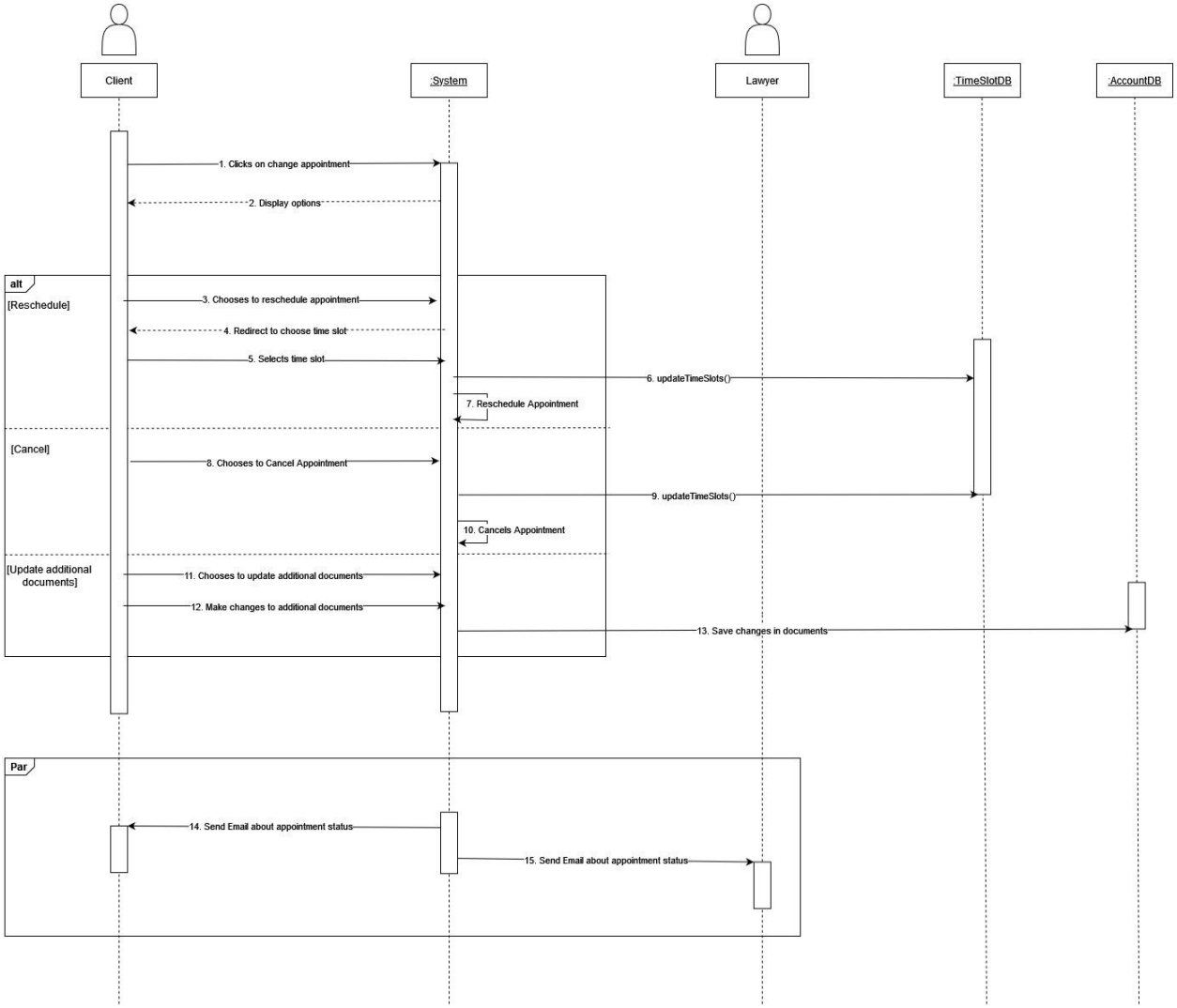
Validate Credentials



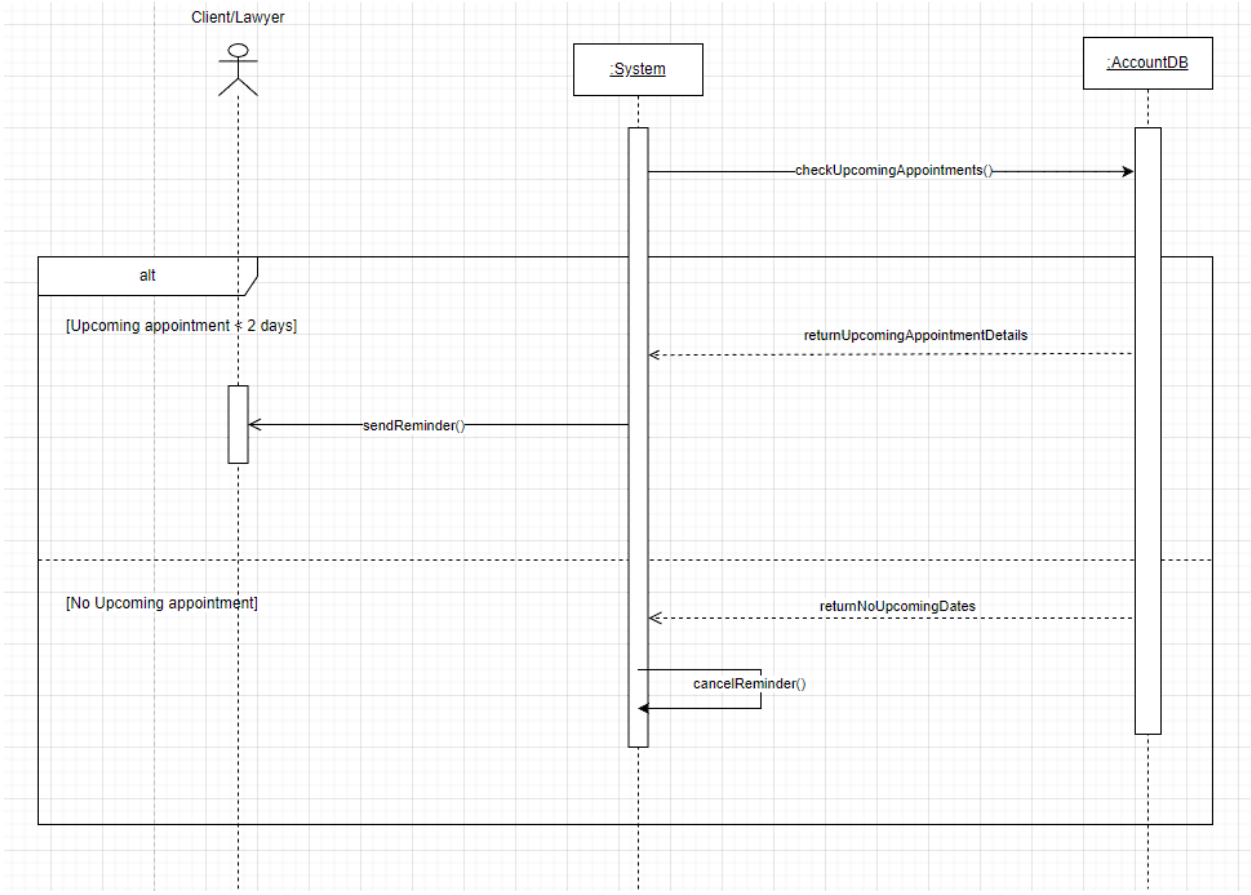
Check Appointment Details



Manage Appointment

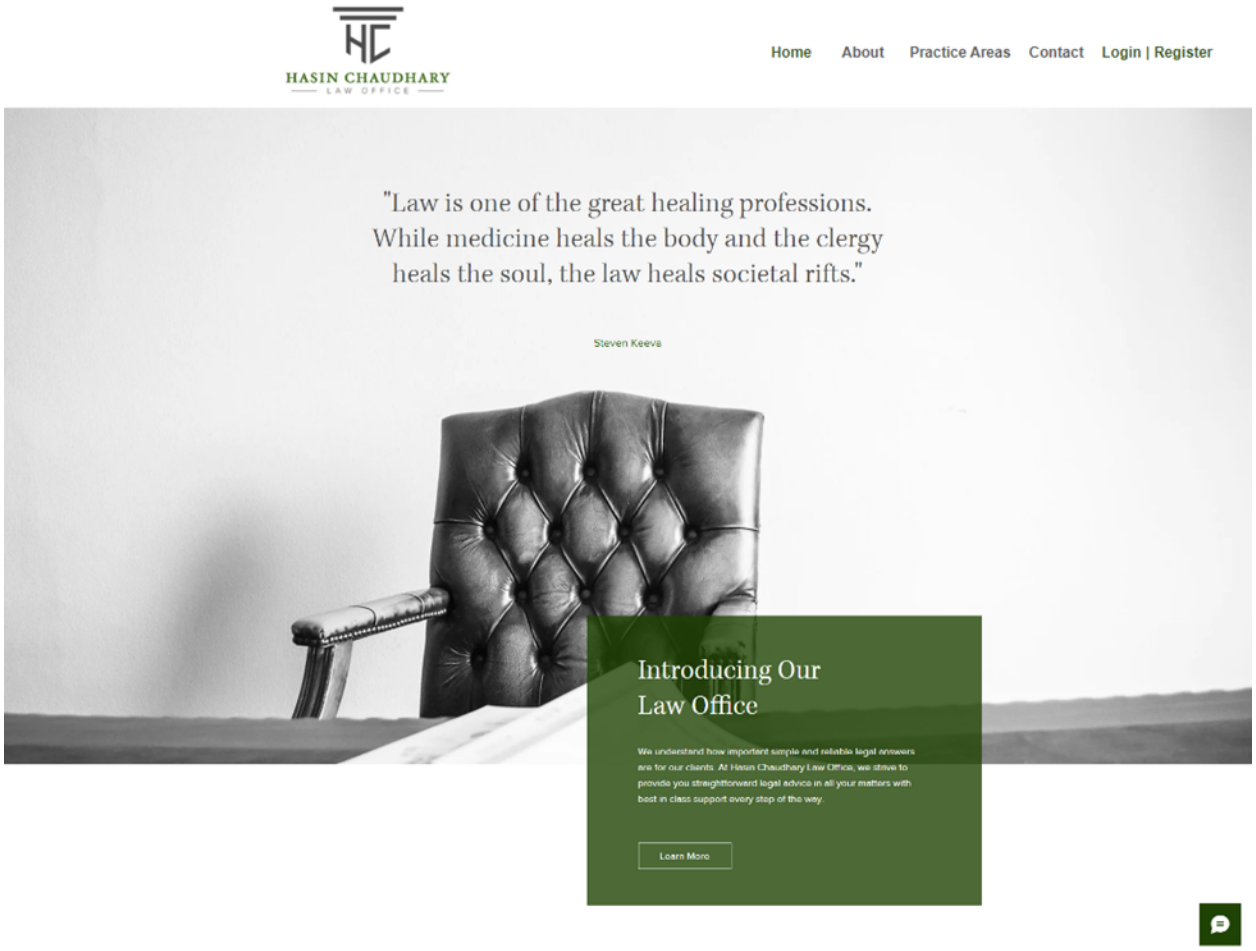


Send Reminder

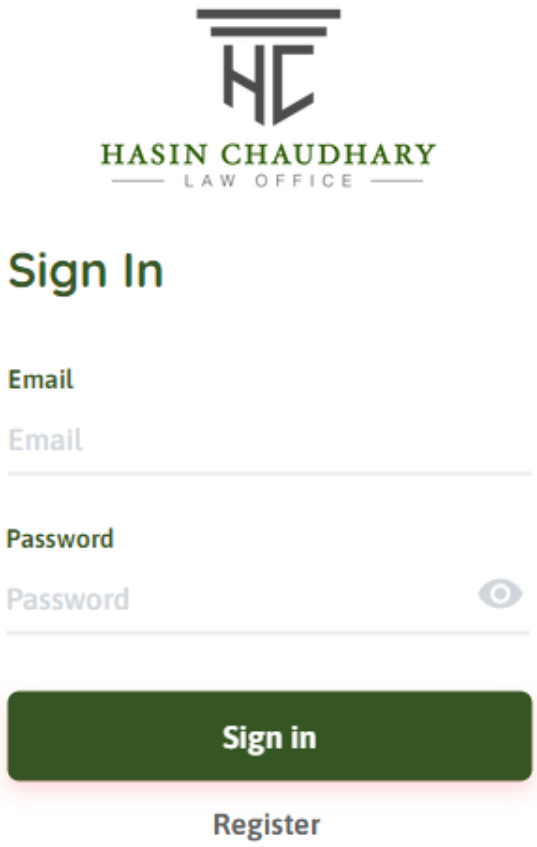


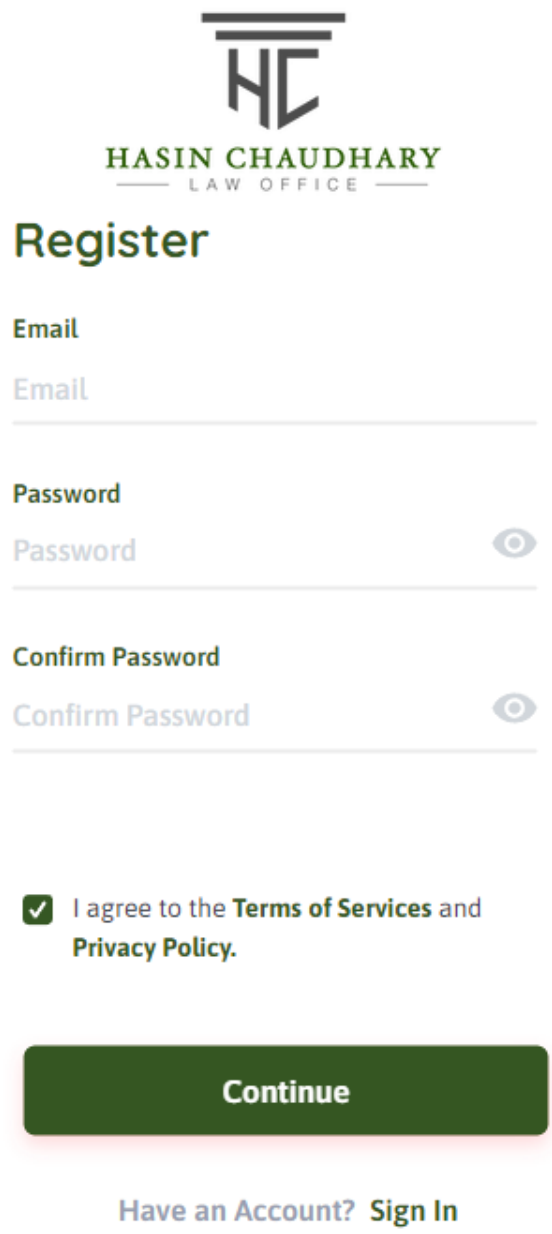
Proof of Concept Implementation Prototype

Home Page

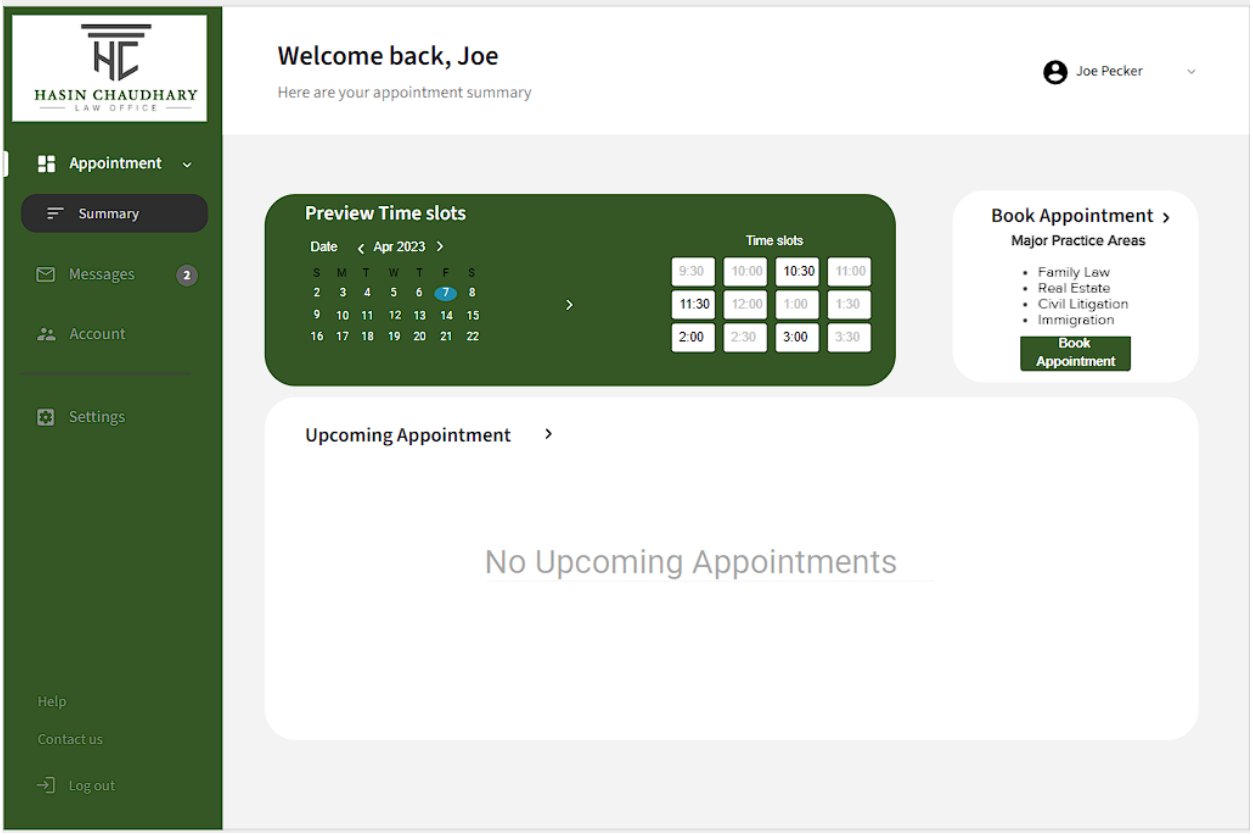


Sign-In & Register





Account Information Page



Intake Form

Please Note: The information in this form, including photo identification, is required by the Law Society of Upper Canada under By-Law 7.1 Client Information. By completion and submission of this form you are consenting to provide us personal information about yourself. Completion and submission of this intake form does not create a Lawyer/Client relationship.

Date:

How did you hear about us?

Type of Photo ID Provided:

Personal Information:

Last Name First Name Middle Name

Home Address Apt No. City, Town or Village

Province Postal Code

Telephone No. (Home) (Business) (Cell) (Fax)

Occupation Employer

E-mail Address

Have court proceedings commenced? Yes ☐ / No ☐ Court File #

What are the names of the other parties involved?

Name	Company	Contact

I hereby authorize Hasin Chaudhary Law Office to communicate with me through the e-mail address provided above and state that the contents of this form are true to the best of my knowledge.

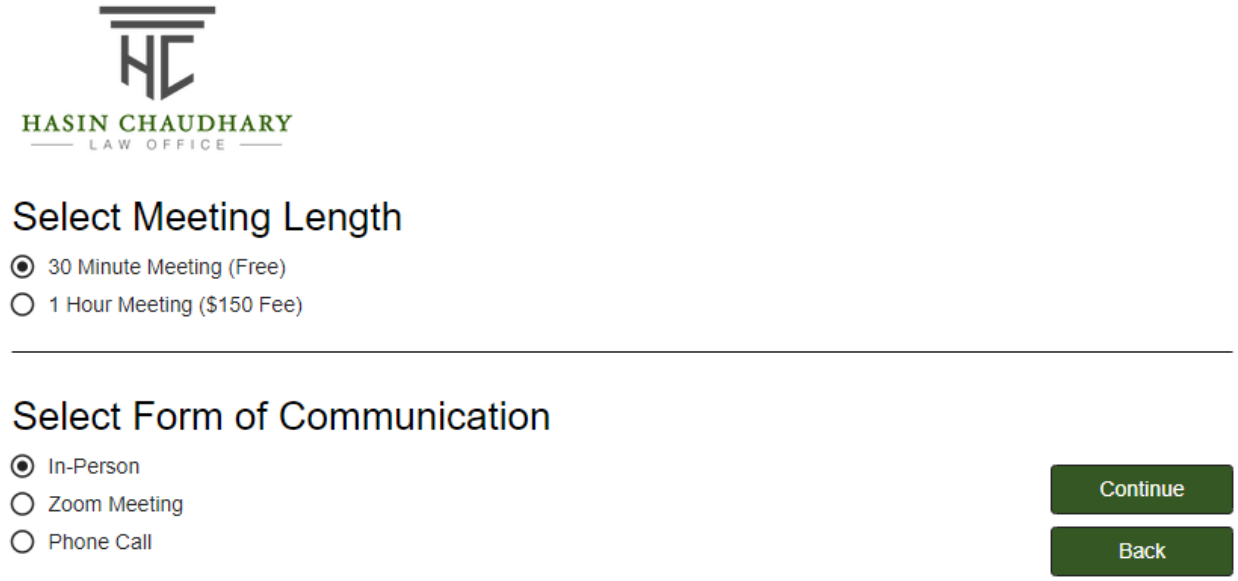
Client Signature: Date:

Add Additional Detail about the Case:

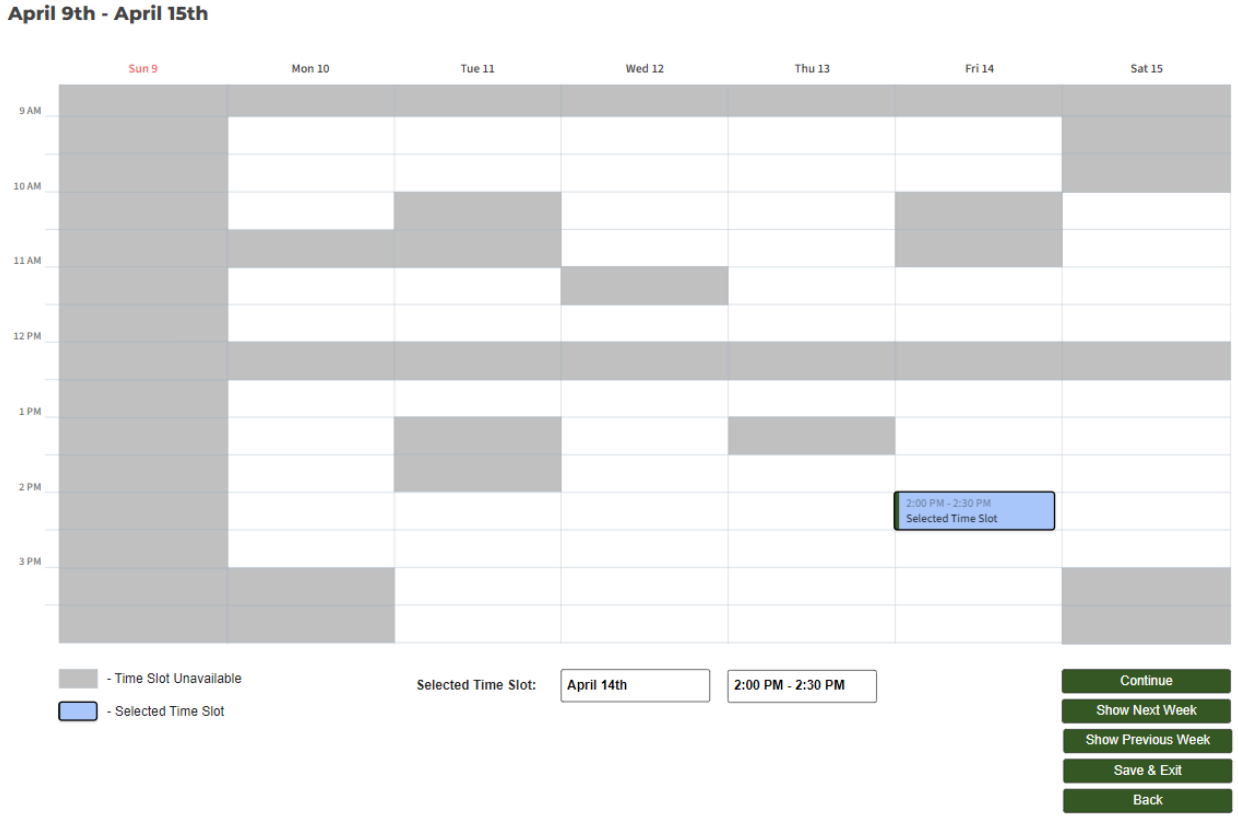
Add Additional Documents

Number of Participants Attending the Meeting:

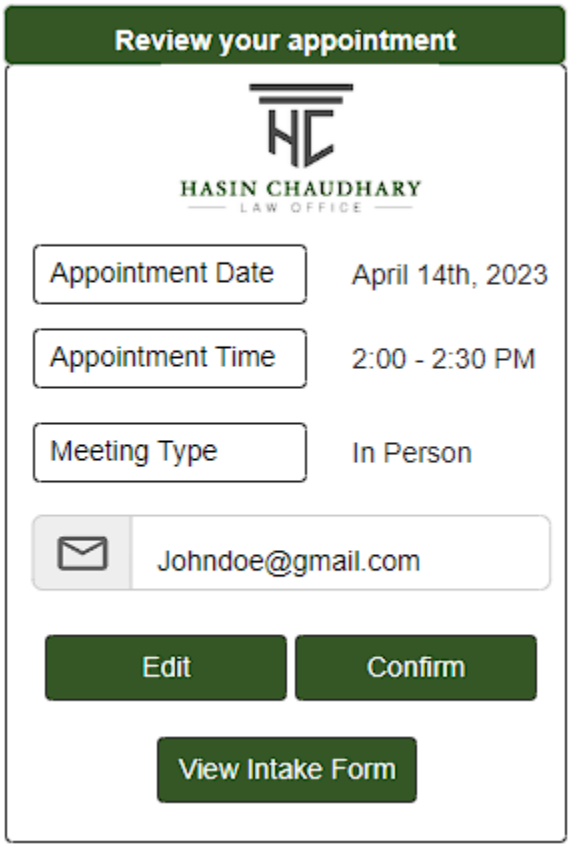
Select Meeting Type



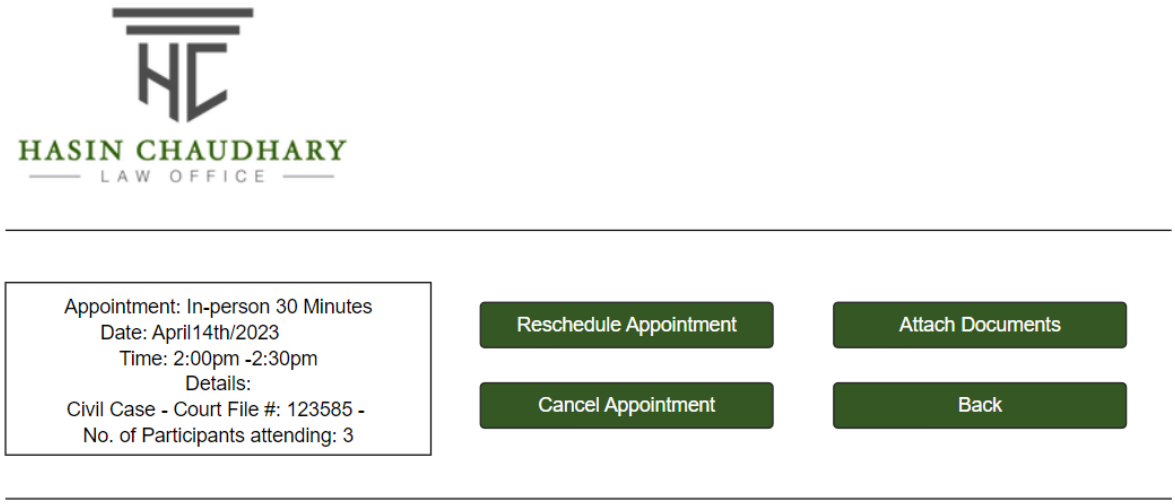
Check Available Time Slots



Review Appointment

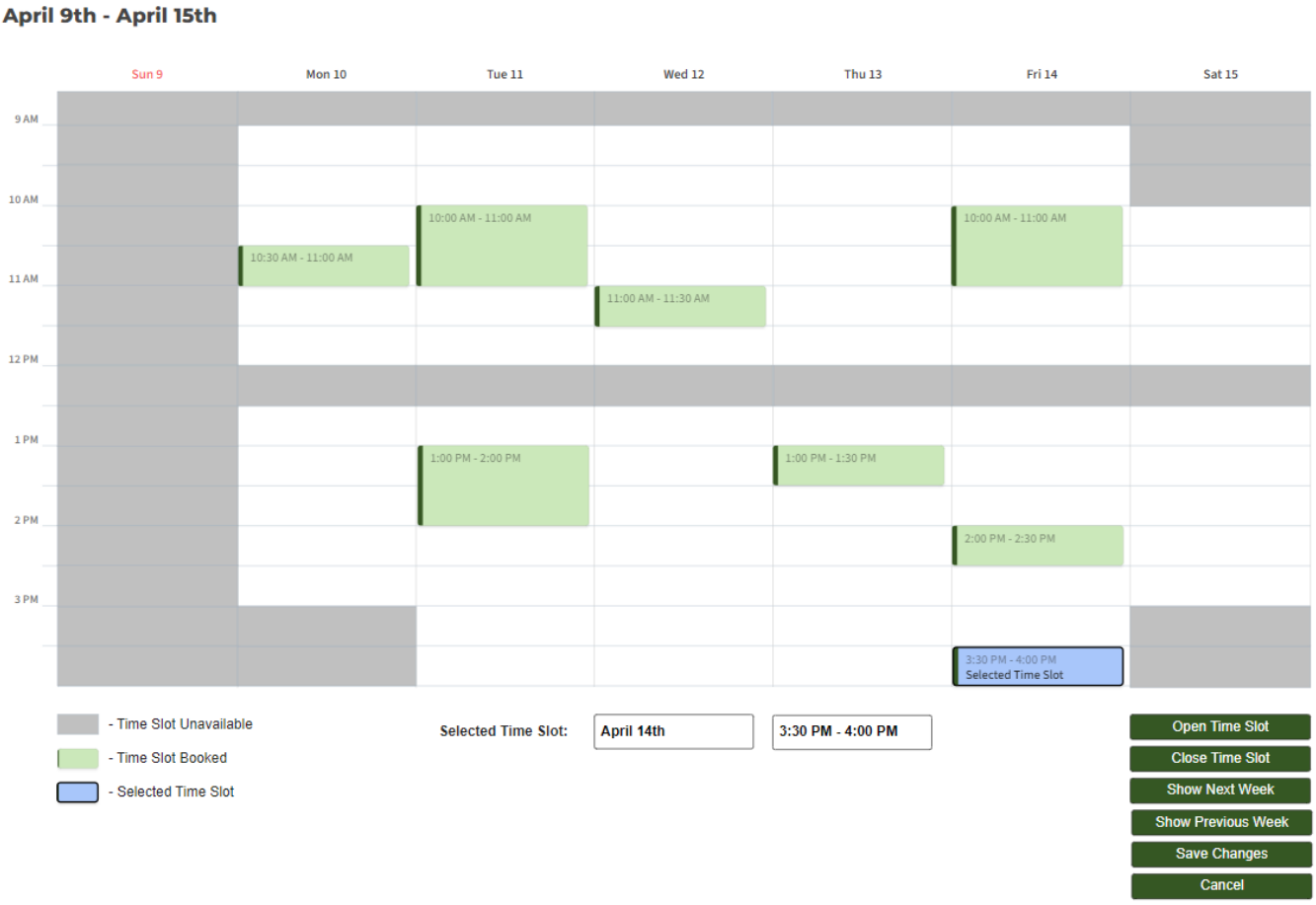


Manage Appointment



Admin Options

Update Time Slots



Implementation and Deployment & Risk Management Plan for the New IS Solution

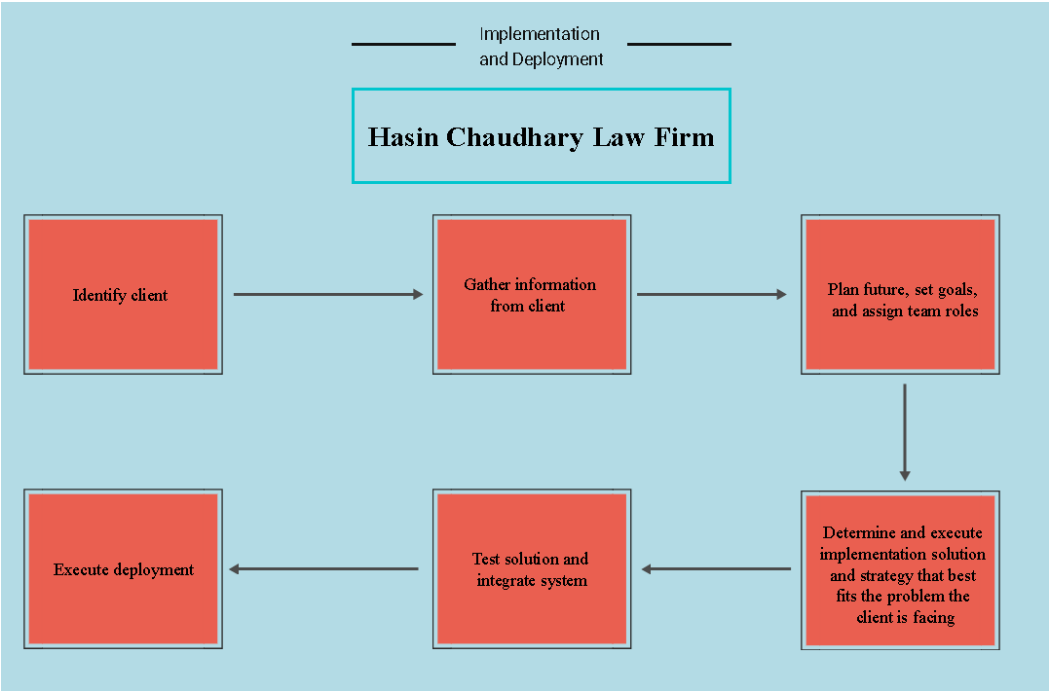
Implementation and Deployment Activities

The purpose of implementing the solution to the problem provides an opportunity for the Hasin Chaudhary law firm to help improve a system in order to apply more efficiency to the current process and system. The implementation and deployment of such a solution refers to the method of turning the design and processes created for the project into a working system as well as deploying it and making it available for users looking to schedule meetings.

Due to the fact that the client wants efficient means for both scheduling and managing appointments, a solution needed to be developed. This would mean that we would have to determine and execute an implementation solution and strategy that best fits the problem that the client is facing. In the process of implementation, it is also essential that the process of the solution would be tested before deployment in order to ensure that the solution works the way it was intended to by integrating the system developed. Lastly in the implementation process, the process of deployment would be executed.

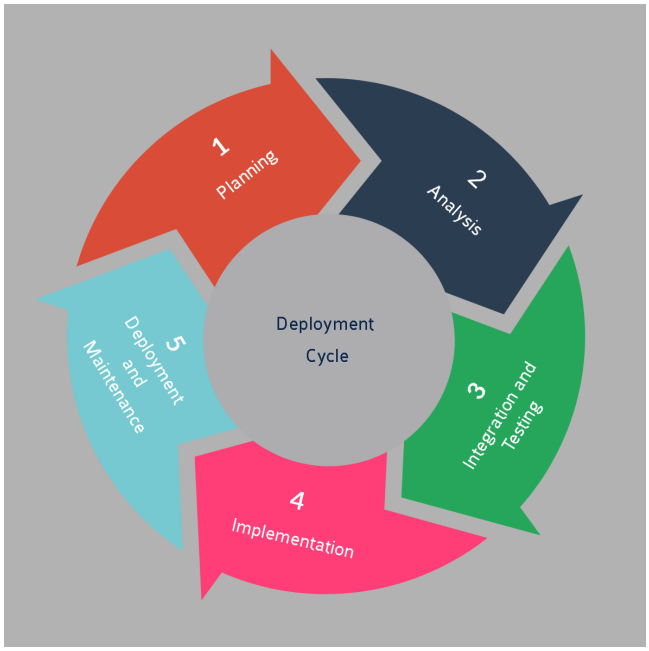
During the meeting with the lawyer (client), our team had gone through the current website and current IT capabilities with one or more glaring problems that were brought to our attention in regards to the current meeting system setup. The problem ultimately had led to a variety of problems for the lawyer and in result faced loss of clients due to issues such as spam emails from bots which led to slow responses from the lawyer and ultimately long response times from both sides due to excessive emails being sent back and forth. In result, our team was tasked to create a solution that would provide more efficiency and improve customer satisfaction. We took this opportunity as a team to improve the current scheduling business process by redesigning the current existing system and improving its performance, effectiveness and overall workflow. In the process of implementation and deployment, our team assigned roles, planned the future, and set goals. After gathering information from the lawyer, we were able to offer a solution for the current website where the clients would be able to book appointments, manage appointments, view and update time slots, as well as providing the lawyer administrator capabilities allowing for more control. The client would have to register or sign-up for an account where they would be led to a homepage where they could manage appointments, view messages, or make any updates to their account or settings. In order for a client to book an appointment, he/she would have to fill out an intake form where personal information and photo identification is required. From there, they would have to select the meeting length and form of communication. After doing so, they would be taken to a new page where they would select an available time slot from a weekly calendar where they would then have to confirm the appointment. They are then presented with information and details in regards to the appointment and case, where the client can reschedule or cancel the appointment. From the point of view of

the administrator (lawyer), they would be able to have a detailed view of the weekly calendar where they can see all booked time slots, where options are available to open and close time slots. Ultimately, this process provides a solution that provides both an efficient means for scheduling and managing appointments between all parties involved by providing a scheduling system on the website that would streamline booking appointments. In result, the implementation of this process will provide many more opportunities for the client and improve the current existing website by introducing the proposed solution.



Deployment plans are necessary because it helps to ensure that the process is ready to be executed so that the problem can have the solution it needs. Ultimately in this process, it is important to know what comes after deployment whether that's offering support for the system or implementing roles and responsibilities for the team after deployment which is why maintenance is key in the process. In the process our team went through, we planned out and analyzed a solution for the client. In result, we came up with the solution of updating the current system by taking advantage of IT services as it is more efficient to create a managing and scheduling experience that is seamless for all parties involved. Rather than wasting precious time with emails which can be counterproductive, time consuming and inefficient, it is important to take advantage of IT and automation services and solutions as it creates for more of a streamlined process which would benefit our client. Before deployment, it is essential for integration and testing to be done before implementation of the process. When it has been confirmed to be a working solution and has been fully tested to be operational, the solution can be fully implemented and deployed. In the last step of the process, it is crucial that maintenance

is involved as it is important that the newly implemented solution would remain working and operational without any issues post-implementation.



Risk Assessment and Mitigation Activities

The risk management plan for our client's appointment booking system will help reduce any risks involving the system that will affect the lawyer (our client) and also any risks that may negatively impact the lawyers potential clients. First, we have to look at any potential risks involving the appointment booking system. One risk we have to access is privacy, the privacy of the lawyer's clients must be kept secure and safe. Each client fills out an intake form when attempting to book an appointment. This form contains personal information regarding the client which is saved to their account. It is due to this form that we take privacy very seriously as it contains sensitive information about the lawyers' clients. To mitigate the risk we have to keep the data secured and encrypted to ensure their personal information is not accessed by any third party hackers. Another risk we have to assess is the risk of overload because of too many people trying to access the webpage. If there are a lot of people trying to book an appointment it may crash the webpage. To mitigate this risk cloud services and scalability will be utilized to help to reduce it. In addition, another error could be if two people book the same time slot at the same time. The system might not register the appointments which can create a mix up for the customer and lawyer. To mitigate this risk a member of the tech team can monitor people booking appointments once every few days to ensure the system is running smoothly. Also, we must ensure that the system is running perfectly with minimal errors. The tech team can set up a button on the website to report any problems the customers find. This can result in helping the tech team identify errors that they did not find on their own and can fix quickly. The mitigation activities to help reduce risks are to monitor the system weekly to ensure it is running smoothly and also to monitor the intake forms to keep them secure and private.